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Therapeutic Exercise

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I repeat my advice to take a great deal of exercise on foot. Health is the first requisite after morality.

—Thomas Jefferson

SUMMARY PEARLS

- Exercise prescriptions should be designed to enhance physical fitness, promote health by reducing risk factors for chronic disease, and ensure safety during exercise participation.
- When prescribing an exercise program, providers should consider its key components: frequency, intensity, time (duration), and type (mode).
- Intensity of exercise is measured generally as mild, moderate, or vigorous. These levels can be calculated in various ways, such as the maximum heart rate method, the heart rate reserve method, or the rating of perceived exertion.
- Time of exercise can be measured by duration of physical activity or by total caloric expenditure, and daily exercise can either be accumulated in one continuous complete session or in multiple sessions of shorter duration.
- In regard to type of exercise, a combination of resistance training, flexibility training, and aerobic exercise is generally recommended for most adults.
- Progression of an exercise regimen should be based on individuals' proficiency with their current regimen and future goals as well as the type (mode) of exercise in consideration.

General Principles

Regular physical activity is an important component of a healthy lifestyle. Increases in physical activity and cardiorespiratory fitness have been shown to reduce the risk for death from not only coronary heart disease but from all causes. The primary focus on achieving these health-related goals in the past has been on prescribing exercise to improve cardiorespiratory fitness, body composition, and strength. Recently, the Centers for Disease Control and Prevention (CDC) and the American College of Sports Medicine (ACSM) suggested that the focus be broadened to address the needs of more sedentary individuals, especially those who cannot or will not engage in structured exercise programs. There is increasing evidence showing that regular participation in moderate-intensity physical activity is associated with health benefits even when aerobic fitness remains unchanged. To reflect this evidence, the CDC and ACSM are now recommending that every adult in the United States accumulates 30 minutes or more of moderate-intensity physical activity on most, and preferably all, days of the week. Those who follow these recommendations can experience many of the health-related benefits of physical activity and, if they are interested, work to achieve higher levels of fitness.[52,53,124,137]

Important in prescribing exercise is an understanding of the principles of specificity and periodization. The principle of specificity states that metabolic responses to exercise occur most specifically in those muscle groups being used. Furthermore, the types of adaptation will be reflective of the mode and intensity of exercise. The principle of periodization reflects the importance of incorporating adequate rest to accompany harder training sessions. Overall training programs (macrocycles) are divided into phases (microcycles), each with specific desired effects (i.e., enhancing a particular energy system or sport-specific goal).

This chapter provides a brief overview of the basic fundamentals of exercise physiology, including the metabolic energy systems, and the basic muscle and cardiorespiratory physiology associated with exercise. It will then provide an overview of the exercise prescription according to the current ACSM guidelines and the fundamentals of exercise programming, including preexercise screening.

Energy Systems

A 70-kg human has an energy expenditure at rest of approximately 1.2 kcal/min, with less than 20% of the resting energy expenditure attributed to skeletal muscle. During intense exercise, however, total energy expenditure can increase 15 to 25 times above resting values, resulting in a caloric expenditure between 18 and 30 kcal/min. Most of this increase is used to provide energy to the exercising muscles that can increase energy requirements by a factor of 200.[31,119]

The energy used to fuel biologic processes comes from the breakdown of adenosine triphosphate (ATP), specifically from the chemical energy stored in the bonds of the last two phosphates of the ATP molecules. When work is performed, the bond between the last two phosphates is broken, producing energy and heat:

$$\text{ATPAT Pase} \rightarrow \text{ADP} + \text{Pi} + \text{energy.}$$

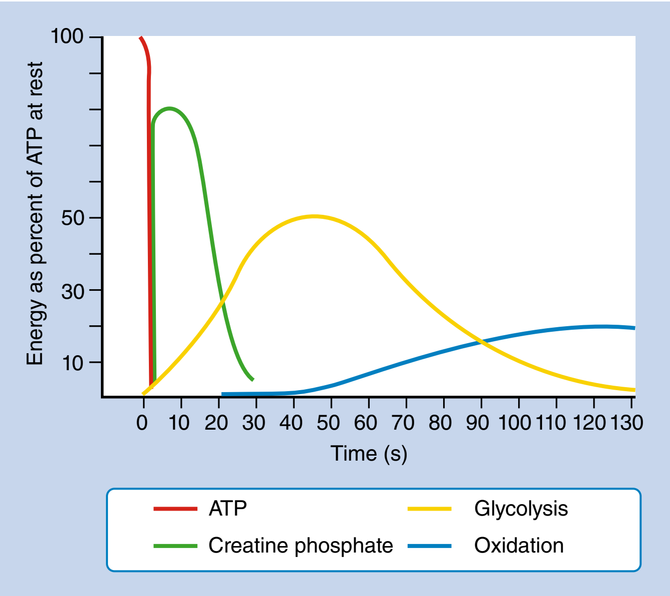
The limited stores of ATP in skeletal muscles can fuel approximately 5 to 10 seconds of high-intensity work (Fig. 16.1). ATP must be continuously resynthesized from adenosine diphosphate (ADP) to allow exercise to continue.[77,130] Muscle fibers contain three metabolic pathways for producing ATP: the creatine phosphate system, rapid glycolysis, and aerobic oxidation.[31,119,124]

Creatine Phosphate System

When limited stores of ATP are nearly depleted during high-intensity exercise (5–10 seconds), the creatine phosphate system transfers a high-energy phosphate from creatine phosphate to rephosphorylate ATP from ADP:

$$\text{ADP} + \text{creatine phosphate creatinekinase} \rightarrow + \text{ATP} + \text{creatine.}$$

Because it involves a single reaction, this system can provide ATP at a very rapid rate. Because there is a limited supply of creatine phosphate in the muscle, however, the amount of ATP that can be produced is also limited. There is enough creatine phosphate stored in skeletal muscle for approximately 25 seconds of high-intensity work (see Fig. 16.1). The ATP–creatine phosphate system lasts approximately 30 seconds (5 seconds for the stored ATP and 25 seconds for creatine phosphate). This provides energy for activities such as sprinting and weightlifting. The creatine phosphate system is considered an anaerobic system because oxygen is not required.[31,119,124]



• **Fig. 16.1** Energy sources in relation to duration of contraction. Muscular metabolism available from the various substrates participating in supplying energy during the first 2 minutes of an attempted maximal contraction. The relative contribution of each substrate at any moment is indicated. The intensity of metabolic activity over the 2-minute period is adjusted to the change of the isometric tension produced during a sustained voluntary maximal contraction. *ATP,* Adenosine triphosphate. (Redrawn from DeLateur BJ. Therapeutic exercise to develop strength and endurance. In: Kottke FJ, Stillwell GK, Lehmann JF, eds. *Krusen's Handbook of Physical Medicine and Rehabilitation.* Saunders; 1982.)

Rapid Glycolysis (Lactic Acid System)

Glycolysis uses carbohydrates primarily in the form of muscle glycogen as a fuel source. When glycolysis is rapid, the pathways that normally use oxygen to make energy are circumvented in favor of other faster (yet less efficient) paths that do not require oxygen. As a result, only a small amount of ATP is produced anaerobically, and lactic acid is produced as a byproduct of the reaction. For many years, lactic acid was considered to be the waste product caused by inadequate oxygen supply. Lactic acid limited physical activity by building up in muscles and leading to fatigue and diminished performance. Since the early 1980s, there has been a fundamental change in thought, and evidence now shows that a limited oxygen supply is not required for lactic acid production. Lactate is produced and used continuously under fully aerobic conditions. This is referred to as the cell-to-cell lactate shuttle, in which lactate serves as a metabolic intermediate tying together glycolysis (as an end product) and oxidative metabolism. Once lactic acid is formed, there are two possible venues it can take. The first involves conversion into pyruvic acid and subsequently into energy (ATP) under aerobic conditions (see Aerobic Oxidation System, later). The second involves hepatic gluconeogenesis using lactate to produce glucose, which is known as the Cori cycle. Anaerobic oxidation starts as soon as high-intensity exercise begins and dominates for approximately 1.5 to 2 minutes (see Fig. 16.1). It would fuel activities such as middle-distance sprints (400-, 600-, and 800-m runs) or events requiring sudden bursts of energy, such as weightlifting.

Although glycolysis is considered an anaerobic pathway, it can readily participate in the aerobic metabolism when oxygen is available and is considered the first step in the aerobic metabolism of carbohydrates.[31,119,124]

Aerobic Oxidation System

The final metabolic pathway for ATP production combines two complex metabolic processes: the Krebs cycle and the electron transport chain. The aerobic oxidation system resides in the mitochondria. It is capable of using carbohydrates, fat, and small amounts of protein to produce energy (ATP) during exercise through the process of oxidative phosphorylation. During exercise, this pathway uses oxygen to completely metabolize the carbohydrates to produce energy (ATP), leaving only carbon dioxide and water as byproducts. The aerobic oxidation system is complex and requires 2 to 3 minutes to adjust to a change in exercise intensity (see Fig. 16.1). It has an almost unlimited ability to regenerate ATP, however, limited only by the amount of fuel and oxygen that is available to the cell. Maximal oxygen consumption, also known as $\dot{V}O_2$ max, is a measure of the power of the aerobic energy system and is generally regarded as the best indicator of aerobic fitness.[31,119,124] All the energy-producing pathways are active during most types of exercise, but different exercise types place greater demands on different pathways. The contribution of the anaerobic pathways (creatine phosphate system and glycolysis) to exercise energy metabolism is inversely related to the duration and intensity of the activity. The shorter and more intense the activity, the greater the contribution of anaerobic energy production, whereas the longer the activity and the lower the intensity, the greater the contribution of aerobic energy production. In general, carbohydrates are used as the primary fuel at the onset of exercise and during high-intensity work. However, during prolonged exercise of low to moderate intensity (>30 minutes), a gradual shift from carbohydrate toward an increasing reliance on fat as a substrate occurs. The greatest amount of fat use occurs at approximately 60% of maximal aerobic capacity.[31,119,124]

Cardiovascular Exercise

The cardiorespiratory system consists of the heart, lungs, and blood vessels. The purpose of this system is the delivery of oxygen and nutrients to the cells and the removal of metabolic waste products to maintain the internal equilibrium.[77,119,124]

Cardiac Function

Heart Rate

Normal resting heart rate (HR) is approximately 60 to 80 beats/min (though it may be as low as 40–50 in highly fit or endurance athletes). HR increases in a linear manner with the work rate and oxygen uptake during exercise. The magnitude of HR response is related to age, body position, fitness, type of activity, the presence of heart disease, medications, blood volume, and environmental factors such as temperature and humidity. HR during maximal exercise can exceed 200 beats/min depending on the individual's age and training state. With the onset of dynamic exercise, HR increases in proportion to the relative workload. The maximal HR (HR_{max}) decreases with age and can be estimated in healthy males and females with the following formula: $HR_{max} = 220 - \text{age}$. There is considerable variability in this estimation for any fixed age, with a standard deviation of ±10 beats/min.[77,119,124]

Stroke Volume

Stroke volume (SV) is the amount of blood ejected from the left ventricle in a single beat. SV is equal to the difference between end-diastolic volume and end-systolic volume. Greater diastolic filling (preload) will increase SV. Factors that resist ventricular outflow (afterload) will result in a reduced SV. SV is greater in males than in females. At rest in the upright position, it generally ranges from 60 to 100 mL/beat, whereas maximal SV approximates 100 to 120 mL/beat. During exercise, SV increases in a curvilinear relationship with the work rate until it reaches near maximal at a level equivalent to approximately 50% of aerobic capacity. SV starts to plateau, and further increases in workload do not result in increased SV primarily because of reduced filling time during diastole.

SV is also affected by body position, with SV being greater in the supine or prone position and lower in the upright position. Static exercise (weight training) can also cause a slight decrease in SV because of increased intrathoracic pressure.[77,99,119,124]

Cardiac Output

Cardiac output (Q) is the amount of blood pumped by the heart each minute. It is calculated by the following formula:

$$Q\,(\text{L/min}) = \text{HR (beats/min)} \times \text{SV (mL/beat)}$$

Resting Q in both trained and sedentary individuals is approximately 4 to 5 L/min, but during exercise the maximal Q can reach 20 L/min. Maximum Q in an individual depends on many factors, including age, posture, body size, presence of cardiac disease, and physical conditioning. During dynamic exercise, Q initially increases with increasing exercise intensity by increases in SV and HR. Increases in Q initially beyond 40% to 50% of $\dot{V}O_2$ max, however, are accounted for only by increases in HR.[77,119,124]

Blood Flow

At rest, 15% to 20% of Q is distributed to the skeletal muscles, with the remainder going to visceral organs, the brain, and the heart. During exercise, 85% to 90% of Q is selectively delivered to working muscles and shunted away from the skin and splanchnic vasculature. Myocardial blood flow can increase four to five times with exercise, whereas blood supply to the brain is maintained at resting levels. The difference between the oxygen content of arterial blood and the oxygen content of venous blood is termed the arteriovenous oxygen difference. It reflects the oxygen extracted from arterial blood by the tissues. The oxygen extraction at rest is approximately 25%, but at maximal exercise the oxygen extraction can reach 75%.[77,119,124]

Venous Return

Venous return is maintained or increased during exercise by the following mechanisms.[77,119,124] Contracting skeletal muscle acts as a "pump" against the various structures that surround it, including deep veins, forcing blood back toward the heart. Smooth muscle around the venules contracts, causing venoconstriction. This increases the pressure on the venous side, maintaining blood flow toward the heart. Diaphragmatic contraction during exercise creates lowered intrathoracic pressure, facilitating blood flow from the abdominal area and lower extremities.

Blood Pressure

Blood pressure is the driving force behind blood flow. Systolic blood pressure (SBP) is the maximal force of the blood against the walls of the arteries when cardiac muscle is contracting (systole). Normal resting SBP is less than 120 mm Hg. Diastolic blood pressure (DBP) is the force of the blood against the walls of the arteries when the heart is relaxing (diastole). Normal resting DBP is less than 80 mm Hg.[119] SBP increases linearly with increasing work intensity, at 8 to 12 mm Hg per metabolic equivalent (MET) (where 1 MET = 3.5 mL O_2/kg/min). Maximum values typically reach 190 to 220 mm Hg. Maximum SBP should not be greater than 260 mm Hg. DBP remains unchanged or only slightly increases with exercise.[77,119] As blood pressure is directly related to Q and peripheral vascular resistance, it provides a noninvasive way to monitor the inotropic performance (pumping capacity) of the heart. Failure of SBP to rise, decreased SBP with increasing work rates, or a significant increase in DBP are all abnormal responses to exercise and indicate either severe exercise intolerance or underlying cardiovascular disease.[77,119,124]

Postural Considerations

In the supine position, gravity has less effect on return of blood to the heart; thus the SBP is lower. When the body is upright, gravity works against the return of blood to the heart; thus SBP increases. DBP does not change significantly with body position in healthy individuals.[51,117,124]

Effects of Arm and Leg Exercises

At similar oxygen consumptions, HR, SBP, and DBP are higher during arm work than during leg work. This is primarily because the total muscle mass in the arms is smaller, and consequently a greater percentage of the available mass is recruited to perform the work. In addition, arm work is less mechanically efficient than leg work.[77,119,124]

Pulmonary Ventilation

Pulmonary ventilation ($\dot{V}e$) is the volume of air exchanged per minute and generally is approximately 6 L/min at rest in an average sedentary adult male. However, at maximal exercise, $\dot{V}e$ increases 15- to 25-fold over resting values. During mild to moderate exercise, $\dot{V}e$ increases primarily by increasing tidal volume, but during vigorous activity it increases by increasing the respiratory rate.[51,117] Increases in $\dot{V}e$ are generally directly proportional to an increase in oxygen consumption $\dot{V}O_2$ and carbon dioxide that is produced. At a critical exercise intensity (usually 47%–64% of the $\dot{V}O_2$ max in healthy untrained individuals and 70%–90% $\dot{V}O_2$ max in highly trained individuals), however, $\dot{V}e$ increases disproportionately relative to the $\dot{V}O_2$ (paralleling an abrupt increase in serum lactate and $\dot{V}O_2$). This is called the anaerobic (ventilatory) threshold.[51,117,124]

Anaerobic Threshold

The anaerobic threshold signifies the onset of metabolic acidosis during exercise and traditionally has been determined by serial measurements of blood lactate. It can be noninvasively determined by assessment of expired gases during exercise testing, specifically $\dot{V}e$ and carbon dioxide production $\dot{V}O_2$. The anaerobic threshold signifies the peak work rate or oxygen consumption at which the energy demands exceed the circulatory ability to sustain aerobic metabolism.[51,117,124]

Maximal Oxygen Consumption

The most widely recognized measure of cardiopulmonary fitness is the aerobic capacity, or $\dot{V}O_2$ max. This variable is defined physiologically as the highest rate of oxygen transport and use that can be achieved at maximal physical exertion. The resting oxygen consumption of an individual (250 mL/min) divided by body weight (70 kg) gives the resting energy requirement, 1 MET (approximately 3.5 mL/kg/min). Multiples of this value are used to quantify levels of energy expenditure. For example, running a 6-mph pace requires 10 times the resting energy expenditure, giving an aerobic cost of 10 METs, or 35 mL/kg/min. Because there is little variation in HR_{max} and maximal systemic arteriovenous oxygen difference with physical training, $\dot{V}O_2$ max virtually defines the pumping capacity of the heart. When expressed as milliliters of oxygen per kilogram of body weight per minute (mL/kg/min) or in METs, it is considered the best index of physical work capacity or cardiorespiratory fitness.[51,117,124]

Oxygen Pulse

The oxygen pulse (mL/beat) is the ratio of $\dot{V}O_2$ (mL/min) to HR (beats/min) when both measures are obtained simultaneously. Oxygen pulse increases with increasing work effort. A low value during exercise indicates an excessive HR for workload and can be an indicator of heart disease.[34]

Respiratory Quotient and Respiratory Exchange Ratio

The respiratory quotient (RQ) is the ratio of CO_2 produced by cellular metabolism to O_2 used by tissues. It quantifies the relative amounts of carbohydrate and fatty acids being oxidized for energy. An RQ of 0.7 implies dependence on free fatty acids. An RQ of 1.0 indicates dependence on carbohydrate. The RQ does not exceed 1.0. The respiratory exchange ratio (RER) reflects pulmonary exchange of CO_2 and O_2 at rest and during exercise. The RER also ranges between 0.7 and 1.0 during rest, and it can reflect substrate preference. During strenuous exercise, however, the RER can exceed 1.0 because of increasing metabolic activity not matched by $\dot{V}O_2$ and additional CO_2 derived from bicarbonate buffering of lactic acid. The terms *respiratory quotient* and *respiratory exchange ratio* are often used interchangeably, but their distinction is important.[34]

Effects of Exercise Training

Cardiovascular System

The effects of regular exercise on cardiovascular activity can be grouped into changes that occur at rest, during submaximal exercise, and during maximal work (Box 16.1).[119,124] Regular exercise can also affect a number of physiologic measurements (Box 16.2).

Detraining

The changes induced by regular exercise training generally are lost after 4 to 8 weeks of detraining. If training is reestablished, the rate at which the training effects occur does not appear to be faster.[119,124]

• BOX 16.1 Effects of Regular Exercise on Cardiovascular Activity

Changes at Rest

- Heart rate decreases, probably secondary to decreased sympathetic tone, increased parasympathetic tone, and a decreased intrinsic firing rate of the sinoatrial node.
- Stroke volume increases secondary to increased myocardial contractility.
- Cardiac output is unchanged at rest.
- Oxygen consumption does not change at rest.[a]

Changes at Submaximal Work[b]

- Heart rate decreases, at any given workload, because of the increased stroke volume and decreased sympathetic drive.
- Stroke volume increases because of increased myocardial contractility.
- Cardiac output does not change significantly because the oxygen requirements for a fixed workload are similar. The same cardiac output is generated, however, with a lower heart rate and higher stroke volume.
- Submaximal oxygen consumption does not change significantly because the oxygen requirement is similar for a fixed workload.
- Arteriovenous oxygen (AVO2) difference increases during submaximal work.
- Lactate levels are decreased because of metabolic efficiency and increased lactate clearance rates.[a]

Changes at Maximal Work

- Maximum heart rate does not change with exercise training.
- Stroke volume increases because of increased contractility and/or increased heart size.
- Maximal cardiac output increases because of increased stroke volume.
- Maximal oxygen consumption ($\dot{V}o_{2max}$) increases primarily because of increased stroke volume.
- Ability of the local mitochondria to use oxygen is improved.[a] However, because measured a-$\dot{V}o_2$ difference represents whole-body a-$\dot{V}o_2$ difference, there is generally little change in the measured a-$\dot{V}o_2$ difference at maximal work.)

[a]From Rupp JC. Exercise physiology. In: Roitman JL, Bibi KW, Thompson WR, eds. *ACSM Health Fitness Certification Review.* Lippincott Williams & Wilkins; 2001; Seto C. Basic principles of exercise training. In: O'Connor F, Sallis R, Wilder R, et al., eds. *Sports Medicine: Just the Facts.* McGraw-Hill; 2005.
[b]Submaximal work is defined as a workload during which a steady state is achieved.

• BOX 16.2 Physiologic Changes After a Regular Exercise Program

Blood Pressure

- In normotensive individuals, regular exercise does not appear to have a significant impact on resting or exercising blood pressure. In individuals with hypertension, there can be a modest reduction in resting blood pressure as a result of regular exercise.[a]

Blood Volume Changes

- Total blood volume increases because of an increased number of red blood cells and expansion of the plasma volume.[a]

Blood Lipids

- Total cholesterol can be decreased in individuals with hypercholesterolemia.
- High-density lipoprotein cholesterol increases with exercise training.
- Low-density lipoprotein cholesterol can remain the same or decrease with regular exercise.
- Triglycerides can decrease in those with elevated triglycerides initially. This change is facilitated by weight loss.[a]

Body Composition

- Total body weight usually decreases with regular exercise.
- Fat-free weight does not normally change.
- Percent body fat declines.[a]

Biochemical Changes

- Stored muscle glycogen increases.
- The percentage of fast- and slow-twitch fibers does not change, but the cross-sectional area occupied by these fibers may change because of selective hypertrophy of either fast- or slow-twitch fibers.[a]

Energy System Changes

- Specificity of training refers to the fact that the changes that occur are specific to the muscles and energy systems that are being used.
- Chronic anaerobic training using the ATP–creatine phosphate system results in improved capacity and power of this system because of enhancement of enzyme activity and increases in the amount of ATP and creatine phosphate in the muscle.
- Anaerobic glycolysis is improved if the training program uses this system, resulting in increased stores of muscle glycogen and improved ability of enzymes in the system.
- Regular aerobic training improves ($\dot{V}o_{2max}$). It increases muscle glycogen and triglyceride stores, as well as the rate at which carbohydrates and fat are metabolized.[a]

ATP, Adenosine triphosphate.

[a]From Rupp JC. Exercise physiology. In: Roitman JL, Bibi KW, Thompson WR, eds. *ACSM Health Fitness Certification Review.* Lippincott Williams & Wilkins; 2001; Seto C. Basic principles of exercise training. In: O'Connor F, Sallis R, Wilder R, et al., eds. *Sports Medicine: Just the Facts.* McGraw-Hill; 2005.

Overtraining

Overtraining fatigue syndrome presents as a prolonged, decreased sport-specific performance, usually lasting greater than 2 weeks. It is characterized by premature fatigability, emotional and mood changes, lack of motivation, infections, and overuse injuries. Recovery is markedly longer and variable among affected athletes, sometimes taking months before the athlete returns to baseline performance (Box 16.3).[119,124]

• BOX 16.3 Symptoms of Overtraining Syndrome

- Sudden decline in quality of work or exercise performance
- Extreme fatigue
- Elevated resting heart rate
- Early onset of blood lactate accumulation
- Altered mood states
- Unexplained weight loss
- Insomnia
- Injuries related to overuse

Exercise Prescriptions

Exercise prescriptions are designed to enhance physical fitness, promote health by reducing risk factors for chronic disease, and ensure safety during exercise participation. The fundamental objective of the prescription is to bring about a change in personal health behavior to include habitual physical activity. The optimal exercise prescription for an individual is determined from an objective evaluation of that individual's response to exercise, including observations of HR, blood pressure, rating of perceived exertion (RPE) to exercise, electrocardiogram when appropriate, and $\dot{V}O_2$ max measured directly or estimated during a graded exercise test. The exercise prescription should be developed with careful consideration of the individual's health status, medications, risk factor profile, behavioral characteristics, personal goals, and exercise preferences.[53,124,137]

Components of an Exercise Prescription

Mode is the particular form or type of exercise. The selection of mode should be based on the desired outcomes, focusing on exercises that are most likely to sustain participation and enjoyment. Intensity is the relative physiologic difficulty of the exercise. Intensity and duration of exercise interact and are inversely related. Duration or time is the length of an exercise session. Frequency refers to the number of exercise sessions per day and per week. Progression (overload) is the increase in activity during exercise training, which, over time, stimulates adaptation.[53,119,124,137] These five essential components apply when developing an exercise prescription for individuals of all ages and fitness levels. Each component of fitness (e.g., cardiorespiratory endurance, muscular strength and endurance, and flexibility) has its own specific exercise prescription associated with it. The following section reviews the ACSM guidelines for each component of fitness.

Exercise Prescription for Cardiorespiratory Endurance

Cardiorespiratory endurance is the ability to take in, deliver, and use oxygen. It is dependent on the function of the cardiorespiratory system (heart and lungs) and the cellular metabolic capacities. The degree of improvement that can be expected in cardiorespiratory fitness is directly related to the frequency, intensity, duration, and mode of exercise. $\dot{V}O_2$ max can increase between 5% and 30% with training. It has become apparent recently, however, that the level of physical activity necessary to achieve the majority of health benefits is less than that needed to attain a high level of cardiorespiratory fitness.[53,124,137]

ACSM Recommendations for Cardiorespiratory Endurance Training

Mode

The best improvements in cardiorespiratory endurance occur when large muscle groups are engaged in rhythmic aerobic activity. Various activities can be incorporated into an exercise program to increase enjoyment and improve compliance. Appropriate activities include walking, jogging, cycling, rowing, stair climbing, aerobic dance ("aerobics"), water exercise, and cross-country skiing as examples.[53,124,137]

Intensity

ACSM recommendations describe minimal exercise intensity as exercise of at least moderate intensity (i.e., 40%–60% of $\dot{V}O_2$ max that noticeably increases HR and breathing). A combination of moderate and vigorous exercise (≥60% of $\dot{V}O_2$ max that results in substantial increases in HR and breathing) is ideal for the attainment of improvements in health and fitness in most adults.[53,124,137] There are a number of ways of calculating intensity, and because of limitations in using $\dot{V}O_2$ calculations for prescribing intensity, the most common methods of setting the intensity of exercise to improve or maintain cardiorespiratory fitness use HR and RPE.[53,115,124] The appropriate exercise intensity is one that is safe and compatible with a long-term active lifestyle for that individual and achieves the desired outcome given the time constraints of the exercise session. The ACSM recommends an intensity that will elicit an RPE within a range of 12 to 16 on the original 6 to 20 Borg scale (Table 16.1).

Heart rate methods. HR is used as a guide to set exercise intensity because of the relatively linear relationship between HR and percentage of $\dot{V}O_2$ max. It is best to measure HR_{max} during a progressive exercise test when possible because HR_{max} declines with age. HR_{max} can be estimated with the following equation: $HR_{max} = 220 - \text{age}$. This estimation has significant variance, with a standard deviation of 10 beats/min.[53,115,124,137]

Maximum heart rate method. One of the oldest methods of setting the target HR range uses a straight percentage of the HR_{max}.

TABLE 16.1 Borg Scale of Perceived Exertion

Level	Perceived Exertion
6	—
7	Very, very light
8	
9	Fairly light
10	
11	
12	
13	Somewhat hard
14	
15	Hard
16	
17	Very hard
18	
19	Very, very hard
20	

From Noble B, Borg G, Jacobs I. A category-ratio perceived exertion scale: relationship to blood and muscle lactates and heart rate. *Med Sci Sports Exerc.* 1983;15:523-528.

Using 70% to 85% of an individual's HR_{max} approximates 55% to 75% of $\dot{V}O_2$ max and provides the stimulus needed to improve or maintain cardiorespiratory fitness.[53,115,124,137] For example, if the HR_{max} is 180 beats/min, then the target HR (70%–85% of HR_{max}) would range from 126 to 153 beats/min (see Chapter 27).

Heart rate reserve method. The HR reserve (HRR) method is also known as the Karvonen method. In this method, the target range is calculated as follows: subtract standing resting HR (HR_{rest}) from HR_{max} to obtain HRR. Calculate 50% and 85% of the HRR. Add each of these values to the HR_{rest} to obtain the target range. Thus the target range is as follows:

$$\text{Lowtarget HR} = [(HR_{max} - HR_{rest}) \times 0.50] + HR_{rest}$$
$$\text{Hightarget HR } [(HR_{max} - HR_{rest}) \times 0.85] + HR_{rest}$$

The small but systematic differences between the two HR methods occur because the percentage HR_{max} is 55% to 75% of $\dot{V}O_2$ max, whereas the percentage HRR_{max} is 60% to 80% of $\dot{V}O_2$ max. Either method can be used to approximate the range of exercise intensities known to increase or maintain cardiorespiratory fitness or $\dot{V}O_2$ max.[53,124,137]

Rating of perceived exertion. The RPE is a subjective grading of how hard individuals feel they are exercising. Use of RPE is considered an adjunct to monitoring HR. It has been shown to be a valuable aid in prescribing exercise for individuals who have difficulty with HR palpation and in cases where the HR response to exercise may have been altered because of a change in medication. The most commonly used scale of perceived exertion is the Borg scale (see Table 16.1). The average RPE range associated with physiologic adaptation to exercise is 13 to 16 ("somewhat hard" to "hard") on the Borg scale category. One should suit the RPE to the individual on a specific mode of exercise and not expect an exact matching of the RPE to a percentage HR_{max} or percentage HRR. It should be used only as a guideline in setting the exercise intensity.[53,108,115,124,137]

Duration

Duration can be measured by duration of physical activity or by the total caloric expenditure. A dose-response relationship exists between the number of calories used during activity and the health and fitness benefits. Physical activity may be continuous or intermittently accumulated during a day through one or more sessions of activity lasting greater than 10 minutes. The ACSM recommends the following quantity of physical activity for most healthy adults: 1000 kcal/wk (or approximately 150 min/wk or 30 min/day). For most adults, larger quantities of exercise (i.e., ≥2000 kcal/wk or ≥250 min/wk or 50–60 min/day) result in greater health benefits.[53,115,124,137]

Frequency

As mentioned previously, activity can be continuous or intermittent and accumulated over the course of a day through more than one session of activity lasting greater than 10 minutes. The ACSM recommends exercise 3 to 5 days per week. Less conditioned people can benefit from lower intensity, shorter duration exercise performed at higher frequencies per day and/or per week.[53,124,137]

Progression

The rate of progression depends on health and fitness status, individual goals, and compliance rates. Frequency, intensity, and/or duration can be increased to provide overload. The ACSM notes a 5- to 10-minute increase every 1 to 2 weeks over the first 4 to 6 weeks of an exercise is reasonable for a healthy adult.[53,124,137]

Medical Clearance for Exercise Training

Exercise training may not be appropriate for everyone. Patients whose adaptive reserves are severely limited by disease processes may be unable to adapt to or benefit from exercise. In this small subpopulation of people with severe or unstable cardiac, respiratory, metabolic, systemic, or musculoskeletal disease, exercise programming can be fatal, injurious, or simply not beneficial, depending on the clinical status and condition of the individual.[54,137] The recommended level of screening before beginning or increasing an exercise program depends on the risk for the individual and the intensity of the planned physical activity. For individuals planning to engage in low- to moderate-intensity activities, the Physical Activities Readiness Questionnaire (PAR-Q) (Box 16.4) should be considered the minimal level of screening. The PAR-Q was designed to identify the small number of adults for whom physical activity might be inappropriate or those who should receive medical advice concerning the most suitable type of activity.[54,124,137]

Preexercise Evaluation

A preexercise evaluation by a physician is more comprehensive and should begin with a detailed patient history.

Patient History

The patient history should include an assessment of current and previous exercise patterns, patient motivations and barriers to exercise, patient beliefs regarding risks and benefits of exercise, preferred types of exercise activity, a review of heart disease risk factors (e.g., family history of heart disease before the age of 50 years, diabetes, hypertension, smoking, hyperlipidemia, sedentary lifestyle, and obesity), physical limitations, current medical problems (e.g., cardiac, pulmonary, musculoskeletal), a history of exercise-induced symptoms (e.g., chest pain, shortness of breath, hives), time and scheduling considerations, social supports for exercise, and current medications.[55,137] It should be noted that, contrary to prior editions, the 11th edition of the *ACSM Guidelines for Exercise Testing and Prescription* no longer recommends exercise testing or any other type of medical examination as a part of preexercise evaluation because there is a current lack of evidence for the clinical value of exercise testing as a screening tool for potential adverse, injurious, or fatal outcomes during exercise in otherwise asymptomatic individuals.

Identification of Those WHO Need an Exercise Stress Test

The clinical utility of exercise testing to guide medical treatment plans has, however, been strongly demonstrated in patients with the following indications: symptoms suggesting myocardial ischemia, acute chest pain in a patient excluded for acute coronary syndrome, recent acute coronary syndrome treated without coronary angiography or incomplete revascularization, known coronary artery disease with worsening symptoms, prior coronary revascularization (≥5 years after coronary artery bypass graft or ≤2 years after percutaneous coronary intervention), valvular heart disease (to assess exercise capacity and need for surgical intervention), certain cardiac arrythmias to assess chronotropic competence, newly diagnosed heart failure or cardiomyopathy, and exercise intolerance and unexplained dyspnea. Exercise stress testing may also be indicated in occupational-specific settings, such as for firefighters, police officers, or pilots, depending on established national occupational guidelines in those fields. Furthermore, it should be noted that although exercise testing does not appear to demonstrate utility as a prescreening tool for exercise participation risk, exercise testing continues to have utility in development of individualized exercise programs as well as in research associated with assessments relating to maximal and submaximal exercise fitness and performance. In these settings, it is important to be mindful of the risks that maximal and submaximal exercise testing may pose to an individual. It is important to understand the major signs and symptoms that are suggestive of cardiopulmonary disease while prescreening for exercise participation (Box 16.5). Absolute and relative contraindications to exercise testing are listed in Box 16.6.

Exercise Preparticipation Screening

In regard to the revised preexercise evaluation guidelines, new ACSM recommendations have evolved toward the utilization of

• BOX 16.4 The Physical Activity Readiness Questionnaire for Everyone (PAR-Q+)

A "yes" answer to any of the questions indicates the necessity of a physician's referral for a preexercise evaluation before the individual begins or increases physical activity.

- Has your doctor ever said that you have a heart condition OR high blood pressure?
- Do you feel pain in your chest at rest, during your daily activities of living, OR when you do physical activity?
- Do you lose balance because of dizziness OR have you lost consciousness in the past 12 months?
- Have you ever been diagnosed with another chronic medical condition (other than heart disease or high blood pressure)?
- Are you currently prescribed medications for a chronic medical condition?
- Do you currently have (or have had within the past 12 months) a bone, joint, or soft tissue (muscle, ligament, tendon) problem that could be made worse by becoming more physically active?
- Has your doctor ever said that you should only do medically supervised physical activity?

Reprinted with permission from the PAR-Q+ Collaboration (www.eparmedx.com) and the current authors of the PAR-Q+ (Dr. Darren Warburton, Dr. Veronica Jamnik, and Dr. Shannon Bredin). https://eparmedx.com/wp-content/uploads/2025/01/PARQPlus2025Fillable.pdf.

• BOX 16.5 Major Symptoms or Signs Suggestive of Cardiopulmonary Disease

- Pain; discomfort (or other anginal equivalent) in the chest, neck, jaw, arms, or other areas that may result from myocardial ischemia; or other recent onset pain of unknown origin
- Shortness of breath at rest or with mild exertion
- Dizziness or syncope
- Orthopnea or paroxysmal nocturnal dyspnea
- Ankle edema
- Palpitations or tachycardia
- Intermittent claudication
- Known heart murmur
- Unusual fatigue or shortness of breath with usual activities

From Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:33.

• **BOX 16.6 Contraindications to Exercise Testing**

Absolute Contraindications

- Acute myocardial infarction within 2 days
- Ongoing unstable angina
- Uncontrolled cardiac arrhythmia with hemodynamic compromise
- Active endocarditis
- Symptomatic, severe aortic stenosis
- Decompensated heart failure
- Acute pulmonary embolism, pulmonary infarction, or deep venous thrombosis
- Acute myocarditis or pericarditis
- Acute aortic dissection
- Physical disability that precludes safe and adequate testing

Relative Contraindications

- Known obstructive left main coronary stenosis
- Moderate to severe aortic stenosis with uncertain relationship to symptoms
- Tachyarrhythmias with uncontrolled ventricular rates
- Acquired advanced or complete heart block
- Recent stroke or ischemic attach
- Mental impairment with limited ability to cooperate
- Resting hypertension with systolic >200 mm Hg or diastolic >110 mm Hg
- Uncorrected medical conditions, such as significant anemia, important electrolyte imbalance, and hyperthyroidism

From Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:116.

preparticipation screening algorithms to make informed exercise participation decisions and plans. For individuals who do not participate in regular exercise and would like to begin, the algorithm for determining whether preparticipatory medical clearance would be necessary is based largely on the presence of underlying comorbidities and the intensity level at which the individual plans to exercise. For individuals who already participate in regular exercise, underlying comorbidities and symptomatology guide whether medical clearance is needed to continue or increase the regimen, with active signs and symptoms suggestive of cardiovascular, metabolic, or renal disease being indications to discontinue exercise until medical reevaluation and clearance have been completed. These new ACSM preparticipation algorithms are summarized in Figs. 16.2 and 16.3.

Muscle Physiology

Each skeletal muscle is made of many muscle fibers, which range in diameter between 10 and 80 μm. Each muscle fiber, in turn, contains hundreds to thousands of myofibrils. Each myofibril comprises approximately 1500 myosin (thick) filaments and 3000 actin (thin) filaments, which are responsible for muscle contraction (Fig. 16.4).[68,136] Myosin and actin filaments partially interdigitate, causing myofibrils to have alternate light and dark bands. The light bands contain only actin filaments and are called I bands (because they are isotropic to polarized light). Dark bands contain myosin as well as the ends of the actin filaments where they overlap the myosin, and are called A bands (because they are anisotropic to polarized light). Small projections, called cross-bridges, protrude from the surface of myosin filaments along their entire length, except in the very center. The interaction between the myosin cross-bridge and the actin filaments results in contraction.[68,136] The ends of actin filaments are attached to Z disks. From the Z disk, actin filaments extend in either direction, interdigitating with the myosin filaments. The Z disk passes from myofibril to myofibril, attaching the myofibrils across the muscle fiber. Thus the entire muscle fiber has light and dark bands, as do individual myofibrils, and thus the striated appearance of the muscle fiber.[68,136] The portion of a myofibril or the whole muscle fiber between two Z disks is called a sarcomere. The myofibrils within the muscle fibers are suspended in a matrix called sarcoplasm. The sarcoplasm contains potassium, magnesium, phosphate, enzymes, and mitochondria. The sarcoplasm also contains the sarcoplasmic reticulum, an extensive endoplasmic reticulum important in the control of muscle contraction.[64,136]

Physiology of Muscle Contraction

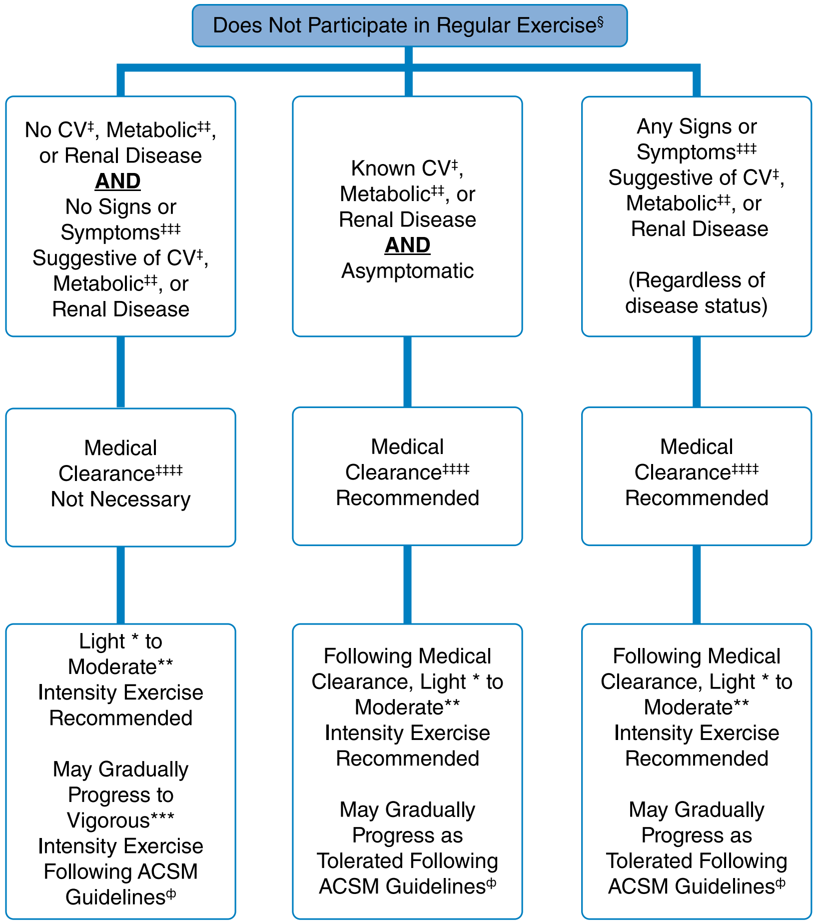
Sliding Filament Mechanism

Muscle contraction occurs by a sliding filament mechanism. In the relaxed state, the ends of actin filaments, derived from two successive Z disks, barely overlap each other while at the same time completely overlapping the myosin filaments. In the contracted state, the actin filaments overlap each other to a great extent, and the Z disks are pulled up to the end of the myosin filaments (Fig. 16.5). The muscle contraction is initiated by the release of acetylcholine from the motor nerve. Acetylcholine opens protein channels in the muscle fiber membrane, allowing sodium to flow into the muscle fiber membrane and initiating a muscle action potential. The action potential depolarizes the muscle fiber membrane, causing the sarcoplasmic reticulum to release calcium. Calcium, in turn, generates attraction between actin and myosin cross-bridges, causing them to slide together.[68,78,136]

Molecular Characteristics of the Contractile Filaments

Each myosin filament is composed of 200 or more myosin molecules. Each myosin molecule is composed of six polypeptide chains: two heavy chains and four light chains. The two heavy chains are wrapped around each other to form a double helix, the tail and arm of the myosin molecule. One end of each of the chains is folded into a globular mass called the myosin head. Therefore two myosin heads are lying side by side. The four light chains are also part of the myosin heads, two to each head (Fig. 16.6). The tails of myosin molecules are bonded together, forming the body of the myosin filament. Protruding from the body, the arm and heads of the myosin molecules are called cross-bridges, which are flexible at two points called hinges. In addition to serving as a component of the cross-bridge, the myosin head also functions as adenosine triphosphatase (ATPase), allowing the head to cleave ATP and energize contraction (Fig. 16.7).

Actin filaments are composed of three protein components: actin, tropomyosin, and troponin (Fig. 16.8). Several G-actin molecules form strands of F-actin. Two F-actin strands are then wound in a double helix. One molecule of ADP is attached to each G-actin molecule. These ADP molecules represent the active sites of the actin filaments, with which myosin cross-bridges interact to cause muscle contraction. Tropomyosin molecules are wrapped around the F-actin helix. In the resting state, tropomyosin covers the active sites of the actin strands, preventing contraction. Attached near one end of each tropomyosin molecule is a troponin molecule. Each troponin molecule consists of three protein subunits. Troponin I has a strong affinity for actin, troponin T for tropomyosin, and troponin C for calcium. In the resting state, the troponin-tropomyosin complex is thought to cover the active sites of actin, inhibiting contraction. In the presence of calcium, this inhibitory effect is removed, allowing contraction to proceed.[68,136]

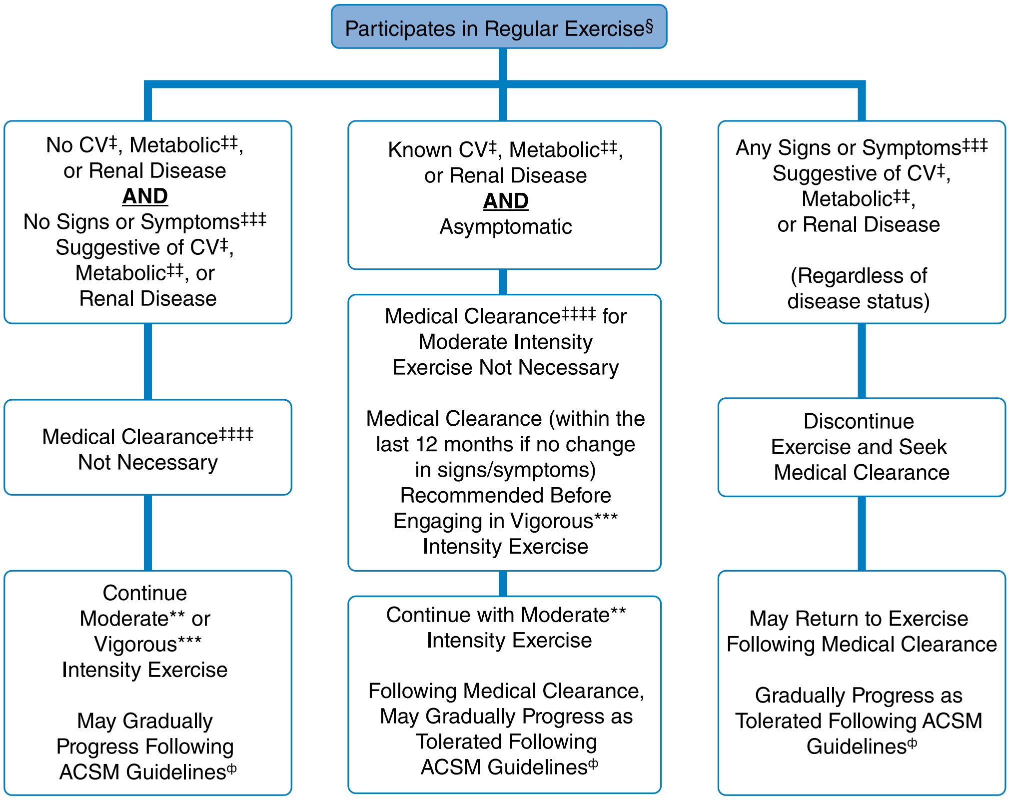


§Exercise Participation	Performing planned, structured physical activity at least 30 min at moderate intensity on at least 3 d • wk^{-1} for at least the last 3 mo
*Light Intensity Exercise	30%–39% HRR or $\dot{V}O_2R$, 2–2.9 METs, RPE 9–11, an intensity that causes slight increases in HR and breathing
**Moderate Intensity Exercise	40%–59% HRR or $\dot{V}O_2R$, 3–5 METs, RPE 12–13, an intensity that causes noticeable increases in HR and breathing
***Vigorous Intensity Exercise	≥60% HRR or $\dot{V}O_2R$, ≥6 METs, RPE ≥14, an intensity that causes substantial increases in HR and breathing
‡Cardiovascular (CV) Disease	Cardiac, peripheral vascular, or cerebrovascular disease
‡‡Metabolic Disease	Type 1 and 2 diabetes mellitus
‡‡‡Signs and Symptoms	At rest or during activity. Includes pain, discomfort in the chest, neck, jaw, arms, or other areas that may result from ischemia; shortness of breath at rest or with mild exertion; dizziness or syncope; orthopnea or paroxysmal nocturnal dyspnea; ankle edema; palpitations or tachycardia; intermittent claudication; known heart murmur; unusual fatigue or shortness of breath with usual activities.
‡‡‡‡Medical Clearance	Approval from a health care professional to engage in exercise
ΦACSM Guidelines	See the most current edition of *ACSM's Guidelines for Exercise Testing and Prescription*

• **Fig. 16.2** Exercise preparticipation screening algorithm for sedentary individuals beginning a new exercise regimen. *HRR,* Heart Rate Reserve; *METs,* Metabolic Equivalents; *RPE,* Rating of Perceived Exertion (Borg Scale). (From Riebe D, Franklin BA, Thompson PD, et al. Updating ACSM's recommendations for exercise preparticipation health screening. Med Sci Sports Exerc. 2015; 47(11):2473–9.)

Muscle Fiber Types

Muscle fibers can be characterized based on their speed of contraction or twitch. Type 1 fibers (slow oxidative) are best suited for endurance activities requiring aerobic metabolism. Type 2 fibers (fast twitch) are most active during activities requiring strength and speed. Type 2 fibers are further categorized into type 2A (fast, oxidative glycolytic) and type 2B (fast glycolytic). Type 2A represents a type of hybrid that retains some oxidative capacity. Features of each muscle type are summarized in Table 16.2. During anaerobic activity, slow oxidative fibers (type 1) are used almost exclusively at intensities below 70% of $\dot{V}O_2$ max. Beyond this intensity, anaerobic pathways are stimulated. At $\dot{V}O_2$ max, both type 1 and type 2 are relied on, and oxygen debt eventually occurs secondary to aerobic metabolism. During isometric contractions, type 2 fibers are generally recruited when force exceeds 20% of the maximal voluntary contraction. If sustained for long



§Exercise Participation	Performing planned, structured physical activity at least 30 min at moderate intensity on at least 3 d • wk^{-1} for at least the last 3 mo
*Light Intensity Exercise	30%–39% HRR or $\dot{V}O_2R$, 2–2.9 METs, RPE 9–11, an intensity that causes slight increases in HR and breathing
**Moderate Intensity Exercise	40%–59% HRR or $\dot{V}O_2R$, 3–5.9 METs, RPE 12–13, an intensity that causes noticeable increases in HR and breathing
***Vigorous Intensity Exercise	≥60%HRR or $\dot{V}O_2R$, ≥6 METs, RPE ≥14, an intensity that causes substantial increases in HR and breathing
‡Cardiovascular (CV) Disease	Cardiac, peripheral vascular, or cerebrovascular disease
‡‡Metabolic Disease	Type 1 and 2 diabetes mellitus
‡‡‡Signs and Symptoms	At rest or during activity. Includes pain, discomfort in the chest, neck, jaw, arms, or other areas that may result from ischemia; shortness of breath at rest or with mild exertion; dizziness or syncope; orthopnea or paroxysmal nocturnal dyspnea; ankle edema; palpitations or tachycardia; intermittent claudication; known heart murmur; unusual fatigue or shortness of breath with usual activities.
‡‡‡‡Medical Clearance	Approval from a health care professional to engage in exercise
ΦACSM Guidelines	See the most current edition of *ACSM's Guidelines for Exercise Testing and Prescription*

• **Fig. 16.3** Exercise participation screening algorithm for individuals already engaged in regular exercise. *HRR,* Heart Rate Reserve; *METs,* Metabolic Equivalents; *RPE,* Rating of Perceived Exertion (Borg Scale). (From Riebe D, Franklin BA, Thompson PD, et al. Updating ACSM's recommendations for exercise preparticipation health screening. Med Sci Sports Exerc. 2015; 47(11):2473–9.)

periods, however, type 2 units can be recruited at thresholds below 20% of maximal voluntary contraction.[29,64,136]

Muscle Fiber Orientation

Parallel muscles have their fibers arranged parallel to the length of the muscle and produce a greater range of motion (ROM) than similar-sized muscles with pennate arrangement. Pennate muscles have shorter fibers that are arranged obliquely to their tendons (similar to the makeup of a feather), increasing cross-sectional area and the power produced.

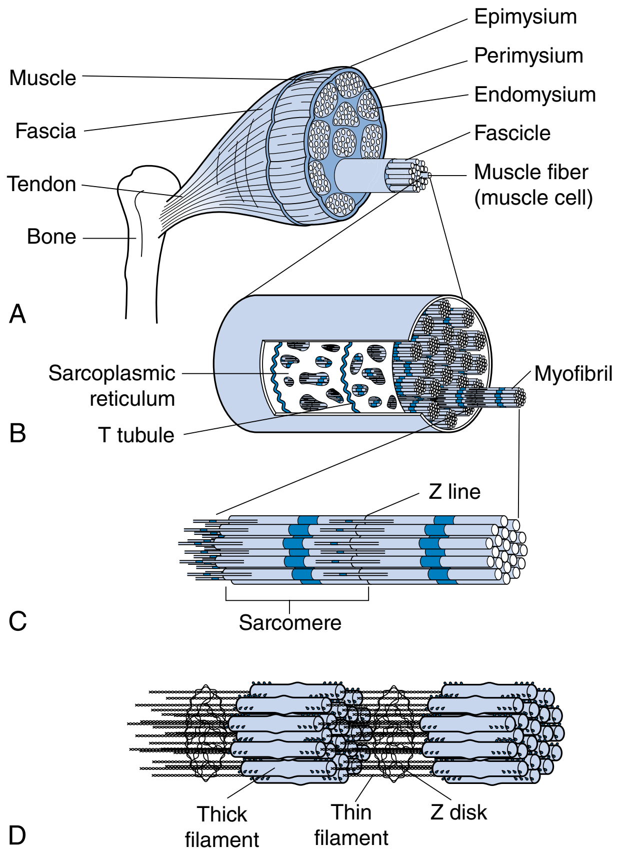
Types of Muscle Contraction

Isometric contractions are contractions in which there is no change in the length of the muscle. No joint or limb motion occurs. Isotonic contractions occur when the muscle changes length, producing limb motion. Concentric contractions occur when the muscle shortens. Eccentric contractions occur when the muscle lengthens. More fast-twitch fibers are recruited during eccentric contractions. Isokinetic contractions occur when muscle contraction is performed at a constant velocity. This can be done only with the assistance of a preset rate-limiting device. This type of exercise does not exist in nature.

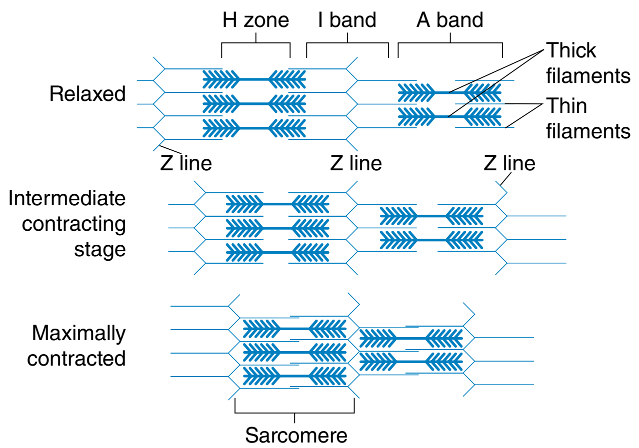
Factors Affecting Muscle Strength and Performance

Determinants of Strength

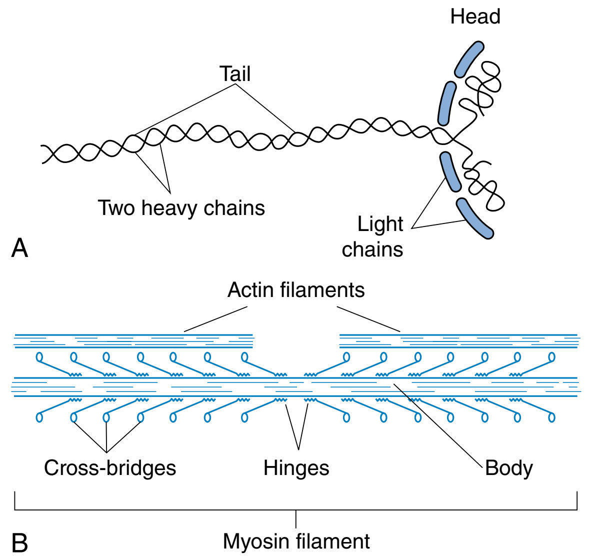
The ability of a muscle to produce force is directly proportional to its cross-sectional area. For parallel muscles, this corresponds to



• **Fig. 16.4** Structure of skeletal muscle. (A) Skeletal muscle organ, composed of bundles of contractile muscle fibers held together by connective tissue. (B) Greater magnification of a single fiber, showing smaller units, myofibrils, in the sarcoplasm. Note the sarcoplasmic reticulum and T tubules forming a three-part structure called a triad. (C) A myofibril magnified further to show sarcomere between successive Z lines. Cross striae are visible. (D) Molecular structure of a myofibril, showing thick myofilaments and thin myofilaments. (Redrawn from Thibodeau GA, Patton KT. *Anatomy and Physiology.* Mosby; 1999.)



• **Fig. 16.5** Sliding filament theory. During contraction, myosin cross-bridges pull the thin filaments toward the center of each sarcomere, thus shortening the myofibril and the entire muscle fiber. (Redrawn from Thibodeau GA, Patton KT. *Anatomy and Physiology.* Mosby; 1999.)



• **Fig. 16.6** The myosin filament. (A) The myosin molecule. (B) The combination of many myosin molecules to form a myosin filament. Also shown are the cross-bridges and the interaction between the heads of the cross-bridges and adjacent actin filaments. (Redrawn from Guyton A. *Textbook of Medical Physiology.* Saunders; 1996.)

the cross section at the bulkiest part of the muscle. For pennate muscles, multiple cross sections are taken at right angles to each of the muscle fibers. Pennate muscles are particularly adapted to force production because many more muscle fibers are contained in pennate muscles, and these fibers are shorter.[87]

Length-Tension Relationship

Maximum force of contraction occurs when a muscle is at its normal resting muscle length. For the muscles, this corresponds to about midrange of joint motion or slightly longer and is the length at which tension just begins to exceed zero. If a muscle is stretched beyond resting length before contraction, resting tension develops, and active tension (the increase in tension during contraction) decreases. Efficiency (percentage of energy that is converted into work instead of heat) occurs at a velocity of contraction of approximately 30% of maximal.[28,29]

Torque-Velocity Relationship

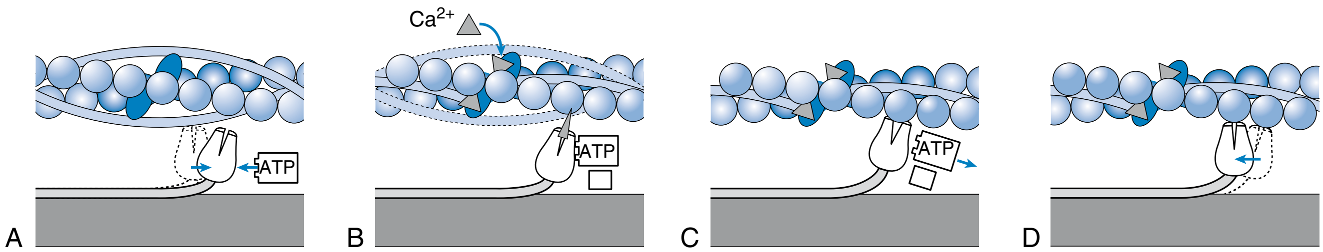
The greatest amount of force is generated by a muscle during fast eccentric (lengthening) contractions.[28,29] The least amount of force is produced during fast concentric (shortening) contractions. The amount of force developed in the order of most force to least force can be summarized as follows: fast eccentric, isometric, slow concentric, and fast concentric (Fig. 16.9).

Effects of Exercise Training

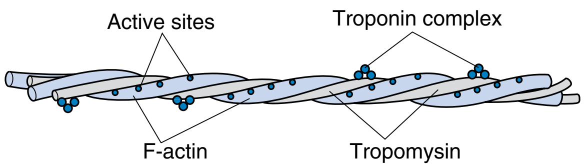
The SAID principle (*s*pecific *a*daptations to *i*mposed *d*emands) states that a muscle will adapt to a specific demand imposed on it, making it better able to handle the greater load.

Neural Adaptations

Observed strength gains within the first few weeks of a weight-lifting program are mostly because of neuromuscular adaptations. The nervous system recruits larger motor units with higher frequencies of stimulation to provide the force necessary to

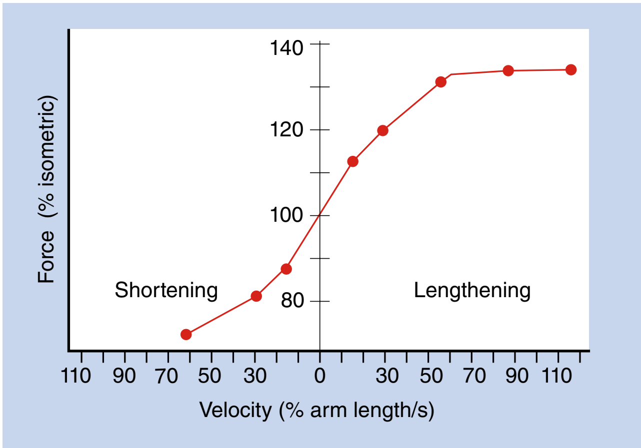


• **Fig. 16.7** The molecular basis of muscle contraction. (A) Each myosin cross-bridge in the thick filament moves into a resting position after an adenosine triphosphate *(ATP)* binds and transfers its energy. (B) Calcium *(Ca^{2+})* ions released from the sarcoplasmic reticulum bind to troponin in the thin filament, allowing tropomyosin to shift from its position, blocking the active sites of actin molecules. (C) Each myosin cross-bridge then binds to an active site on a thin filament, displacing the remnants of ATP hydrolysis: adenosine diphosphate and inorganic phosphate. (D) The release of stored energy from step (A) provides the force needed for each cross-bridge to move back to its original position, pulling actin along with it. Each cross-bridge will remain bound to actin until another ATP binds to it and pulls it back to its resting position (A). (Redrawn from Thibodeau GA, Patton KT. *Anatomy and Physiology.* Mosby; 1999.)



• **Fig. 16.8** Tropomyosin molecules. The actin filament, composed of two helical strands of F-actin and tropomyosin molecules that fit loosely in the grooves between the actin strands. Attached to one end of each tropomyosin molecule is a troponin complex that initiates contraction. (Redrawn from Guyton A. *Textbook of Medical Physiology.* Saunders; 1996.)

overcome the imposed resistance. Early strength gains and increased muscle tension production from training therefore result from a more efficient neural recruitment process. This means that most of the improvement in strength-related functional activities gained on inpatient rehabilitation units is attributable to neural recruitment rather than muscle hypertrophy because of the relatively short length of stays.[91]



• **Fig. 16.9** Relationship of maximal force of human elbow flexor muscles to velocity of contraction. Velocity on the abscissa is designated as a percentage of arm length per second. (Redrawn from Knuttgen HG. Development of muscular strength and endurance. In: Knuttgen HG, ed. *Neuromuscular Mechanisms for Therapeutic and Conditioning Exercises.* University Park Press; 1976.)

TABLE 16.2 Skeletal Muscle Fiber Characteristics

	Type 1 (Slow Oxidative)	Type 2B (Fast Glycolytic)	Type 2A (Fast Oxidative Glycolytic)
Major source of adenosine triphosphate	Oxidative phosphorylation	Glycolysis	Oxidative phosphorylation
Mitochondria	High	Low	High
Myoglobin content	High	Low	High
Capillarity	High	Low	High
Muscle color	Red	White	Red
Glycogen content	Low	High	Intermediate
Glycolytic enzyme activity	Low	High	Intermediate
Myosin adenosine triphosphatase activity	Low	High	High
Speed of contraction	Slow	Fast	Fast
Rate of fatigue	Slow	Fast	Intermediate
Muscle fiber diameter	Small	Large	Intermediate

Muscle Hypertrophy

Muscle hypertrophy represents enlargement of total muscle mass and cross-sectional area. Muscle hypertrophy is more common in fast-twitch than in slow-twitch muscles. Type 2A fibers exhibit the greatest growth, more so than type 2B and type 1 fibers. Muscle hypertrophy is typically experienced after 6 to 7 weeks of resistance training.[33,73] Conversely, muscle atrophy resulting from disuse occurs primarily in type 2 fibers. Virtually all muscle hypertrophy occurs from hypertrophy of the individual muscle fibers. During muscle hypertrophy, the rate of muscle contractile protein synthesis is greater than decay, leading to greater numbers of actin and myosin filaments in the myofibrils. The myofibrils within each muscle fiber split, resulting in more myofibrils in each muscle fiber. Only under very rare conditions of extreme muscle force generation do the numbers of muscle fibers increase (fiber hyperplasia)—and even then by only a few percent.[29,68] Another type of muscle hypertrophy occurs when muscles are stretched to a greater than normal length, causing new sarcomeres to be added at the ends of muscle fibers where they attach to the tendons. Conversely, when a muscle remains shortened at less than its resting length, sarcomeres at the end of the muscle fibers disappear.[27,29]

Exercise Prescription

Advancing in a training program can include increasing the amount of weight lifted (progressive resistive exercise), increasing repetitions, or increasing the velocity of training. A commonly used term to express an individual's current strength level is the one-repetition maximum (1-RM), which is the most weight that can be lifted one time. The current strength fitness level for a particular exercise can be expressed in terms of an individual's multiple RM (e.g., 10-RM is the amount of weight that an individual is capable of lifting 10 times).

Progressive Resistance Exercise

Popular protocols include the DeLorme, Oxford, and daily adjusted progressive resistance exercise (DAPRE) methods.

DeLorme

In the DeLorme method, three sets are performed for each exercise. Ten repetitions are performed in each set. The weight for the first set is 50% of the 10-RM; the second set, 75% of the 10-RM; and the third set, 100% of the 10-RM.[30] This is usually referred to clinically as progressive resistive exercise.

Oxford

In the Oxford technique, an individual starts with 10 repetitions at 100% of the 10-RM, then 10 repetitions at 75% of the 10-RM, then a third set of 10 repetitions at 50% of the 10-RM. This is usually referred to in the clinical setting as regressive resistive exercise.

Daily Adjusted Progressive Resistance Exercise

The DAPRE method of strength training guides the athlete through four sets of exercise per muscle group. The first set in DAPRE involves 10 repetitions at 50% of the athlete's predetermined 6-RM. The second set consists of six repetitions at 75% of the athlete's 6-RM. The third set consists of as many repetitions as can be performed at the athlete's 6-RM. The number of repetitions performed in the third set determines the resistance for the fourth set. If five to seven repetitions were performed in the third set, resistance stays the same. If fewer than five repetitions were performed in the third set, the weight is lowered by 5 pounds. If more than seven repetitions are performed in the third set, the weight is raised by 5 pounds. The fourth set consists of as many repetitions as can be performed to fatigue. The working weight for the next day is established based on the athlete's performance during the fourth set, with the same formula for determining resistance change from the third to the fourth set.[72,90]

It is important to note that none of these three types of resistive training programs have been determined to be superior to the others.

Increasing Number of Repetitions or Rate

Critical to developing muscle strength is to exercise to the point of fatigue. Both high weight–low repetition and low weight–high repetition programs can be effective in achieving strength as long as exercise proceeds to muscle fatigue. Far more repetitions must be carried out with low weights to reach this point of fatigue. In fact, the amount of mechanical work required by low weight–high repetition exercise to reach muscle fatigue might be much greater than during higher weight exercise. Although circumstances exist where low weight–high repetition exercise can be appropriate for training (such as during injury or when training for a highly repetitive task), in general the use of higher weights to fatigue is a more effective means of strength training. Hellebrandt and Houtz[74] advocate increasing the rate of contraction (controlled by a metronome) each session while lifting the same number of repetitions (10–20).[29]

Effects of Aging

Although it was previously believed that strength training in older adults was attributable only to learning or neural factors,[105] reports have also demonstrated that the muscles of older individuals can demonstrate hypertrophy after strength training.[56,87]

ACSM Guidelines for Prescription of Strength-Training Exercise

Muscular strength and endurance can be developed with both dynamic and static exercise. Both forms have their indications, but for most individuals dynamic exercise is recommended. Strength exercise should be rhythmic, performed at low to moderate speed, and performed through a full ROM. Normal breathing should be maintained. Heavy resistance training associated with breath holding can result in dramatic rises in SBP and DBP. A specific technique for each exercise should be closely followed. Both the lifting (concentric) and lowering (eccentric) phases of resistance exercise should be performed in a controlled manner. When possible, training with a partner can provide feedback, assistance, motivation, and safety. Recommended guidelines for strength training are listed in Table 16.3.

Flexibility

Flexibility generally describes the ROM commonly present in a joint or group of joints that allows normal and unimpaired function.[17] More specifically, flexibility has been defined as "the total achievable excursion (within limits of pain) of a body part through its range of motion."[120] With regard to flexibility, the following generalizations can be made. Flexibility is an individually variable, joint-specific, inherited characteristic that decreases with age; varies by sex and ethnic group; bears little relationship

TABLE 16.3 **Recommended Guidelines for Resistance Training**

FITT	Recommendation
Frequency	For novice trainers, each major muscle group should be trained at least 2 days weekly. For experienced exercisers, frequency is secondary to training volume, thus individuals can choose a weekly frequency per muscle group based on personal preference.
Intensity	For novices, 60%–70% 1-rep-max, performed form 8–12 repetitions are recommended to improve muscular fitness. For experienced exercisers, a wide range of intensities and repetitions are effective dependent on the specific muscular fitness goals.
Type	Multiple-joint exercises affecting more than one muscle group and targeting agonist and antagonist muscle groups are recommended for all adults. Single-joint and core exercises may also be included in a resistance training program, typically after performing multiple-joint exercises for that particular muscle group. A variety of exercise equipment and/or body weight can be used to perform these exercises.
Time	For untrained individuals, low volume protocols (<4 sets weekly) can result in significant improvements in muscular fitness. For experienced exercisers, muscle hypertrophy demonstrates a dose-response relationship with number of weekly sets. Although there is certainly and inevitably an upper limit at which this dose-response relationship plateaus or results in regression of muscular fitness, this limit is unique to each individual, and a quantifiable generalizable limit has not yet been conclusively identified.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:153.

with body proportion or limb length; and most important for the purposes of this chapter, can be acquired through training.[22,59,65-67,141] From a developmental standpoint, flexibility is greatest during infancy and early childhood. Flexibility reaches minimal levels between 10 and 12 years of age. It then improves again toward early adulthood, but not sufficiently to allow ranges of motion seen in childhood.[38] The early adolescent growth spurt results in short-term tightness of the joints, probably as a result of increased tension in the connective tissue. Young females are generally more flexible than young males,[11,89,111,118] and this advantage probably persists into adulthood. Achieving a maximal functional ROM is an important goal of many therapeutic exercise regimens. Most typically, increased ROM is achieved via the process of stretching. The term *stretching* defines an activity that applies a deforming force along the rotational or translational planes of motion of a joint.[120] Stretching should respect the lines of geometry of the joint as well as its planes of stability. In addition to stretching, mobilization is used to maintain flexibility. Mobilization moves a joint through its ROM without applying a deforming force.

Importance of Flexibility

Flexibility has only recently been identified as an important component of therapeutic exercise. Cureton[26] emphasized flexibility as an important component of physical fitness after working with swimmers during the 1932 Olympic Games. Later, Kraus[93] stressed the importance of flexibility in preventing low back pain. His work inspired much of the subsequent research on flexibility. It was not until 1964 that Fleishman[49] proved flexibility to be an independent factor in physical fitness that was unrelated to other factors, including strength, power, endurance, and coordination. During the same decade and in another landmark work, DeVries and Housh[37] proved the value of passive stretching in improving flexibility and ROM. Subsequent studies have demonstrated the value of flexibility training for patients in the industrial and athletic settings, for patients with back pain, and for patients following orthopedic surgical procedures. Subsequent to Kraus's work, biomechanical studies have shown that lower limb flexibility is needed for the prevention of lumbar spine injuries.[47] An increased frequency of spondylolysis and spondylolisthesis in individuals with severe hamstring inflexibility has been reported.[110] Cady et al.[14] demonstrated an inverse relationship between flexibility and the incidence of back injuries and workers' compensation costs in a cohort study of firefighters placed on a fitness program.

The work of Salter et al.[88,122,143] emphasized the importance of maintaining motion and flexibility postoperatively in patients who have undergone orthopedic procedures. The benefits of the mobilization of postoperative joints to the surrounding ligamentous and musculotendinous structures have been well established.

In the realm of athletic training, flexibility has been both extensively applied and extensively studied. The proposed benefits of flexibility to athletes include injury prevention, reduced muscle soreness, skill enhancement, and muscle relaxation.[a] With regard to injury prevention, muscles possessing greater extensibility are less likely to be overstretched during athletic activity, lessening the possibility of injury. A 1987 review of all studies of soccer injuries suggested an important role for flexibility in the prevention of injury, and attributed up to 11% of all injuries to poor flexibility.[86] A prospective study of a flexibility program in soccer players demonstrated a correlation between improvement in ROM and reduction in incidence of muscle tears.[45] There is some evidence that delayed muscular soreness can be prevented and treated by static stretching.[35,36a] Flexibility has generally been hypothesized to improve athletic performance through skill enhancement. For example, mastery of the serve in tennis requires sufficient shoulder flexibility. Similarly, proficient golf skills require flexibility throughout the hips, trunk, and shoulders.[21] On the biomechanical level, prestretching a muscle has been shown in several studies to enhance the force of muscle contraction.[8,9,16,17] There is, however, considerable uncertainty regarding two of the most important proposed benefits of flexibility training for athletes: prevention of injury and improvement of performance. Although

[a]References 11, 21, 27, 39, 41, 45, 60, 81, 84, 92.

currently held teaching states that stretching is a preventive measure for athletic injury, it has been pointed out that little conclusive epidemiologic evidence supports this idea.[70,73] In fact, it has been proposed that a certain degree of tightness might protect against injury by allowing load sharing when joints are stressed.[73] Hypermobility or excessive stretching could theoretically result in increased stress on the ligaments, bone, and cartilage at the joint, resulting in injury or arthritis.[65,66,107] In support of this is the fact that there is general agreement that the major predictive factor for joint injury is a previous joint injury or indeed the presence of excessive joint laxity, rather than inadequate flexibility.[b] Presently, the role of preexercise stretching in the prevention of sports-related injury is unclear. A systematic review performed by Thacker et al.[135] for the ACSM failed to find "sufficient evidence to endorse or discontinue routine stretching before or after exercise to prevent injury among competitive or recreational athletes."

With regard to athletic performance, several laboratories have shown that among runners, less flexible individuals have a lower rate of oxygen consumption while covering the same distance at the same speed as their more flexible cohorts.[23] In addition, the aforementioned improvement in contraction strength resulting from prestretching has not been consistently observed in the world of athletics. In fact, it has been shown repeatedly that passive stretching can result in an acute loss of strength.[c] Along the same lines, a recent study of elite female soccer players demonstrated that static stretching before sprinting resulted in worsened performance.[123] Prior stretching does appear to have one reproducible benefit with regard to performance, however: maintenance of strength with the muscle in a lengthened position during and after eccentric exercise.[102] This benefit might be important in resisting injurious muscle elongation during continued sport performance.[102c] The athletic literature seems to show in general that flexibility training, when used appropriately, plays a positive role in sports injury and performance. However, excessive flexibility can actually be both a risk factor for injury and a detriment to performance. Stiff structures appear to benefit from stretching, whereas hypermobile structures require stabilization rather than additional mobilization.

Determinants of Flexibility

The determinants of joint mobility can be subdivided into static and dynamic factors. Static factors include the types of tissues involved, the types and state of collagen subunits in the tissue, the presence or absence of inflammation, and the temperature of the tissue. Dynamic factors include neuromuscular variables such as voluntary muscle control and the length-tension "thermostat" of the musculotendinous unit, as well as external factors such as pain associated with injury.[120]

Static factors.

The most important tissue with regard to flexibility is the muscle-tendon unit, which is the primary target of flexibility training.[120] This structure includes the full length of the muscle and its supporting tissue, the musculotendinous junction, and the full length of the tendon to the tendon-bone junction. Within the muscle-tendon unit, it is the muscle that has the largest capacity for percent lengthening[80,131,132] of the tissues involved in a stretch. A ratio of 95% to 5% for the muscle:tendon length change has been demonstrated.[132] From a mechanical standpoint, muscle is composed of contractile and elastic elements arranged in parallel.[68] Muscle can respond to an applied force or stretch with permanent elongation. Animal studies have shown that this results from an increase in the number of sarcomeres, which translates to increased peak tension of a muscle at longer resting lengths. By contrast, muscle at rest tends to shorten because of its contractile element. This shortening can be permanent and is associated with a reduction in sarcomeres.[63,65,142] Tendon has a much more limited capacity for lengthening than muscle probably because of its proteoglycan content and collagen cross-links (2%–3% of its length, compared with 20% for muscle).[131,132,145] Of the external static factors, temperature has been studied the most. Warmer tissues are generally more distensible than cold ones.[42,140,141]

Dynamic factors.

Perhaps the most clinically and physiologically significant dynamic determinant of flexibility is the muscle length-tension thermostat or feedback control system. Intrafusal fibers (muscle spindles), innervated by gamma motor neurons, lie in parallel with extrafusal contractile fibers. The intrafusal fibers serve the purpose of regulating the tension and length of the muscle as a whole. Muscle spindle length and tension are regulated by the gamma motor neuron, which in turn is subject to influences from the central nervous system (CNS). These include segmental input at the spinal cord level and suprasegmental input from the cerebellum and cortex. Consequently, muscle length and tension can be subject to multiple influences simultaneously.

An additional complicating factor is that receptors in the musculotendinous unit, called the Golgi tendon organs, act to inhibit muscle contraction at the point of critical stresses to the structure. The Golgi tendon organs allow lengthening and facilitate relaxation. When acting in conjunction, these dynamic mechanisms facilitate a response to a stretch in the following way. As the muscle spindle is initially stretched, it sends impulses to the spinal cord that result in reflex muscle contraction. If the stretch is maintained longer than 6 seconds, the Golgi tendon organ fires, causing relaxation.[120]

The relative contribution of static muscle factors and dynamic neural factors to flexibility remains somewhat controversial. It seems clear that the changes in flexibility noted immediately after the institution of a stretching program occur too rapidly to be attributable solely to structural alteration of the muscle and connective tissue. The consensus view is that neural factors probably play the major role in this early flexibility. After prolonged periods of training, changes in sarcomere number can play a role in the establishment of a new elongated muscle length.[120]

Assessment of Flexibility

Flexibility is generally assessed in terms of joint ROM. Joint ROM, in turn, is generally assessed with a goniometer or similar device. A goniometer consists of a 180-degree protractor designed for easy application to joints. The methods used with a goniometer, as well as the normal ranges of motion encountered with these methods, are well standardized.[44,114] Interobserver and intraobserver reliability are good.[41] Limitations of the standard goniometer include application to only single joints at a given time, static measurements only, and difficulty of application to certain joints (e.g., costoclavicular). The Leighton flexometer contains a rotating circular dial marked in degrees and a pointer counterbalanced to remain vertical. It can be strapped to a body segment,

[b]References 44, 45, 62, 85, 87, 125.
[c]References 24, 25, 46, 50, 93, 100.

and ROM is determined with respect to the perpendicular. Its reliability is good but is not fully equivalent to that of the standard goniometer.[71]

The electrogoniometer substitutes a potentiometer for a protractor. The potentiometer provides an electrical signal that is directly proportional to the angle of the joint. This device is able to give continuous recordings during a variety of activities, allowing a more realistic assessment of functional flexibility and dynamic ROM during actual physical activity.

With regard to measuring trunk flexibility, goniometric devices are generally considered inadequate. The Schober test, originally designed to measure spinal flexion and extension in patients with ankylosing spondylitis, is commonly used, as modified by Moll and Wright.[104] Two marks are made along the proximal and distal ends of the lumbar spine, and tape measurements are made between them with the spine in flexion, neutral, and extension. This test has been shown to be more reliable than other methods, including fingertip to floor measurements and the inclinometer technique of Loebl. “Eyeball” measurements show marked variability. These tests of trunk flexibility are all nonspecific, and each is limited to a gross measurement of compound motion of the entire thoracolumbar spine. None of these methods can assess articular mobility in the translational and rotational planes.[120] The optimal measurement of trunk flexibility is probably obtained with plain films, but these have the obvious disadvantages of cost and radiation exposure.

Methods of Stretching

It is important to take several factors into consideration when using a stretching program. Prevention of injury and treatment of specific joint injury, as well as the presence and effects of pain or muscle spasm, require modification of the program. Stretching can be dangerous and might result in significant injury if performed incorrectly.[121,124,126] As with any form of therapeutic exercise, flexibility training must be approached within a program aimed at addressing the specific functional needs of the individual. Numerous options now abound for improving flexibility with stretching techniques. A distinct superiority of any one method has not been demonstrated. For the purposes of this chapter, stretching techniques are divided into the following four categories: ballistic, passive, static, and neuromuscular facilitation.

Ballistic

Ballistic stretching uses the repetitive rapid application of force in a bouncing or jerking maneuver. Momentum carries the body part through the range of motion until the muscles are stretched to their limits. This method is less efficient than other methods because muscle will contract under these stresses to protect itself from overstretching. Additionally, the rapid increase in force can cause injury.[125,133] An example would be the 10-count bouncing toe touches popularized in the 1970s, which has since been abandoned because of lack of efficacy and risk for injury.

Passive

Passive stretching is done by a partner or therapist who applies a stretch to a relaxed joint or extremity. This method requires excellent communication and the slow and sensitive application of force. This method is most appropriately and most safely used in the training room or in a physical or occupational therapy context. Outside these contexts, passive stretching can be dangerous for recreational or competitive athletes because of an increased risk for injury.

Static

Static stretching applies a steady force for 15 to 60 seconds. This method is the easiest and probably the safest type of stretching. Static stretching seems to be particularly helpful as a warmup to any other form of therapeutic or recreational exercise, including athletic activity. Static stretching has the added advantage here of being associated with decreased muscle soreness after exercise.

Neuromuscular Facilitation

The efficacy of stretching afforded by neuromuscular facilitation techniques has been documented in several studies.[121,134] These methods typically require a trained therapist, aide, or trainer. The specific activities most frequently used include hold-relax and contract-relax techniques, characterized by an isometric or concentric contraction of the musculotendinous unit followed by a passive or static stretch. The prestretch contraction is thought to facilitate relaxation and flexibility via the muscle length-tension thermostat (see earlier).

ACSM Guidelines for Prescription of Exercise for Musculoskeletal Flexibility

Optimal musculoskeletal function requires that adequate ROM be maintained in all joints. Particularly important is maintenance of flexibility in the low back and posterior thigh muscles. Poor flexibility in these regions can predispose to low back pain.[88,137]

Some common stretching exercises might not be appropriate for all people, particularly those with a previous injury, joint insufficiency, or other conditions that could place them at risk for injury. Furthermore, exercises requiring substantial flexibility or skill are not recommended for older, less flexible, or less experienced individuals.[82]

Recommended guidelines for flexibility training are listed in Table 16.4.[88,137]

TABLE 16.4 Recommended Guidelines for Flexibility Training

FITT	Recommendations
Frequency	>3 days/week, with daily being most effective.
Intensity	Stretch to the point of feeling tightness or slight discomfort.
Time	Holding a static stretch for 10–30 seconds is recommended for most adults. In older individuals, holding a stretch for 30–60 seconds may confer greater benefit. For proprioceptive neuromuscular facilitation (PNF) stretching, a 3- to 6-second light-moderate contraction (20%–75% of maximum) followed by a 10- to 30-second assisted stretch is desirable.
Type	A series of flexibility exercises for each of the major muscle-tendon units is recommended. Static flexibility, dynamic flexibility, ballistic flexibility, and PNF are each effective.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:160.

Plyometrics

Plyometrics is a relatively recent addition to the panoply of therapeutic exercise. This class of exercises is used primarily in the training of athletes. Proponents of plyometrics advocate it because of its apparent muscle strengthening and injury prevention effects. Plyometric exercises are generally defined as brief, explosive maneuvers that consist of an eccentric muscle contraction followed immediately by a concentric contraction. An example is the action of planting and jumping during sport activity. Here, the process of planting the feet and flexing the hips, knees, and ankles while loading the lower extremities (eccentric contraction) is followed by a quick changeover to concentric contractions while these joints are extended to propel upward into a jump. This type of stretch-shortening cycle is analogous to a spring coiling and uncoiling. Plyometrics allows the body to store elastic energy briefly in the muscle during the eccentric phase. This stored energy, combined with activation of the myotatic stretch reflex, results in a more powerful concentric contraction than is otherwise possible. This type of relatively complex action relies more heavily on the interplay between CNS and muscular system than do many other forms of exercise. Feedback from the CNS to the muscles influences the length of each muscle at any point during the movement, as well as the tension required for maintaining postural stability and initiating or stopping movement.[20] With training, according to proponents of plyometrics, this neuromuscular interplay can be finely tuned. The widespread use of plyometric training in the athletic community suggests general acceptance of these methods by trainers, therapists, and athletes. However, many techniques in use have not been adequately studied. Results of research so far have generally been promising.

Hewett et al.[76] have reported that plyometric jump training improved lower body strength in high school–age females. Specifically, hamstring isokinetic strength and vertical jump height were improved after a 6-week program. A 22% decrease in peak ground reaction forces and a 50% decrease in the abduction-adduction moments at the knee during landing were also observed. In a later study using the same plyometric program, Hewett et al.[75] prospectively analyzed the effect of this neuromuscular training on the incidence of serious knee injuries in female athletes. The authors reported a statistically significant decrease in the number of knee injuries sustained by the trained group vs. the matched control group. Plyometric exercises vary in intensity from simple, two-footed, in-place jumps to hopping and bounding for maximum distance, to depth jumps from boxes of varying heights. Plyometrics has been shown to result in ground reaction forces of four to seven times the body weight.[7,144] Clearly, these exercises should be approached with caution and begun at an elementary level. Progression to more advanced exercises should be based on the individual's proficiency with the basic movements, taking into account baseline levels of strength, stability, and coordination.

Proprioception

Proprioception denotes the process by which information about the position and movement of body parts is related to the CNS. Proprioceptive organs, including muscle (particularly intrafusal spindle fibers), skin, ligaments, and joint capsules, generate afferent information that is crucial to the effective and safe performance of motor tasks. The process of proprioception is unfortunately subject to impairment from injury and disease. For example, knee and ankle ligament injuries have been shown to reduce proprioception. The same is true for both osteoarthritis and rheumatoid arthritis.[6,48] Neuropathies, most notably diabetic neuropathy, can also cause significant loss of proprioception.[127] Proprioception has also been shown to decrease with age.[128] The importance of proprioception to injury prevention and rehabilitation from injury is generally accepted. Impaired proprioception has been associated with an increased risk for joint damage, athletic injury, and falls. Decreased joint proprioception is thought to influence the progressive joint deterioration associated with osteoarthritis, rheumatoid arthritis, and Charcot disease.[5,6] In a study of soccer players, a significantly greater incidence of ankle injury was observed among players with abnormal proprioceptive testing results as compared with those who tested within normal measurements.[139] Some findings also suggest that return to sport after knee injury might be more dependent on proprioception than on ligament tension.[6] It has also been demonstrated in several studies that the risk for falling in the older adult population correlates with postural sway, a variable that is determined in large part by proprioception.[94,96,97,138]

Proprioceptive exercise regimens, by definition, seek to improve joint and limb position sense. These exercises are typically used after an injury has occurred to a joint that has resulted in a deficit in proprioception. For example, the tilt or wobble board is commonly used after ankle ligamentous injuries. Classically, the unidirectional boards are used first, with a progression to multidirectional boards. This type of training has led to measurably improved position sense in athletes.[58] Other proprioceptive exercises include carioca (sideways running) and backward walking or running. It has also been shown that elastic bandaging improves position sense in individuals with previously impaired proprioception,[6] perhaps through stimulation of proprioceptors in the skin.

Neurofacilitation Techniques

CNS dysfunction poses a unique set of challenges to both the patient and the treatment team. The following therapeutic exercise techniques were developed specifically for patients with CNS impairment, particularly impairment resulting from an acquired cortical lesion (i.e., stroke or brain injury).

Proprioceptive Neuromuscular Facilitation

This form of therapy uses resistance to indirectly facilitate movement. The therapist provides maximal resistance to the stronger motor components of specific spiral and diagonal movement patterns, thereby facilitating the weaker components of the patterns. Proprioceptive neuromuscular facilitation techniques are best applied to patients with hypotonia associated with supraspinal lesions to promote normalization of tone. In patients with spasticity, these techniques can actually further increase tone in a potentially detrimental manner.[82,83]

Brunnstrom Techniques

The Brunnstrom techniques use resistance and primitive postural reactions to facilitate gross synergistic movement patterns and increase muscle tone during early recovery from CNS injury.[10] During later stages, Brunnstrom techniques emphasize development of isolated movement and control. Similar to proprioceptive neuromuscular facilitation, this approach is thought to be effective in normalizing tone in a patient with hypotonic or flaccid hemiplegia.

Bobath Techniques

The neurodevelopmental techniques developed by Berta Bobath and Karel Bobath differ significantly from proprioceptive neuromuscular facilitation and Brunnstrom methods. The Bobath techniques use reflex inhibitory movement patterns to inhibit increased tone. These inhibitory patterns, which are generally antagonistic to the primitive synergistic patterns, are performed without resistance. Neurodevelopmental techniques also incorporate advanced postural reactions to stimulate recovery. Advocates of these techniques claim reduction of hypertonicity and facilitation of motor recovery as their primary benefits.[57] The techniques described in this section are all in common clinical use, with most therapists using an eclectic approach, borrowing some from each. There is no convincing evidence, however, that any of these methods actually alter the natural history of recovery from neurologic insult. These approaches to therapy seem to be most useful in providing compensatory techniques during the course of recovery. Using these methods, patients are able to improve performance in and gain independence with such tasks as making transfers, stretching, bed mobility, and safe ambulation.[57]

Exercise for Special Populations

Physical Inactivity and Obesity

Physical inactivity is associated with increased fat and visceral adipose tissue accumulation. Not only does physical inactivity contribute to development of heart disease, but it also increases chances of breast and colon cancer, depression, diabetes, and metabolic syndrome. Sedentary jobs have significantly increased in recent years, and physical activity–demanding jobs constitute only 25% of the workforce. Regular activity is protective by regulating weight and improving the body's use of insulin. Being active is beneficial for blood pressure, blood lipid levels, glucose levels, clotting factors, the health of blood vessels, and inflammation, which is a powerful promoter of cardiovascular disease. Only 48% of adults meet the 2008 ACSM Physical Activity Guidelines. Based on age-adjusted estimates, non-Hispanic Black (41.1%) and Hispanic (42.2%) adults were more inclined to be inactive compared with non-Hispanic White adults. People who are insufficiently physically active have a 20% to 30% increased risk of all-cause mortality compared with those who engage in at least 30 minutes of moderate-intensity physical activity most days of the week.[61]

Although most of the comorbidities relating obesity to coronary artery disease increase as body mass index (BMI) increases, they also relate to body fat distribution. Visceral fat causes central adiposity and increases risk of insulin resistance and diabetes mellitus, heart disease, stroke, and dementia. Long-term longitudinal studies, however, indicate that obesity not only relates to but can also independently predict coronary atherosclerosis. This holds true for both males and females with minimal increases in BMI. In a 14-year prospective study, middle-aged females with a BMI greater than 23 but less than 25 had a 50% increase in risk of nonfatal or fatal coronary heart disease,[98] and males aged 40 to 65 years with a BMI greater than 25 but less than 29 had a 72% increased risk.[116] It has been shown that weight loss also shortens the QT interval, thus further reducing the risk of heart disease.[15]

Treatment of obesity should be tailored according to severity and the presence of comorbidities, such as congestive heart failure, hypertension, dyslipidemia, noninsulin-dependent diabetes, and obstructive sleep apnea.[43]

Exercise for Fat Reduction

Successful weight loss requires a combination of diet and exercise. The threshold for change in body weight appears to be 30 minutes of exercise per day, with more marked losses noted with 60 min/day. Moderate-intensity, land-based activity such as cycling, brisk walking, or pushing a lawn mower is safe for most adults and will reduce the risk of coronary heart disease by approximately 30%. The aim should be to reach 60% to 85% of resting HR during exercise. If exercising for an extended period is not possible, it may be broken into periods of 10 minutes at a time of continuous moderate-vigorous activity. This regimen should be performed with major muscle group activation (all extremities, chest, abdomen, back) at least 2 days per week.[18] Swimming in the absence of caloric restriction did not appear to result in weight loss. In fact, individuals gained weight, although this appeared to be all muscle weight, thus increasing lean body mass. It stands to reason, however, that swimming can be effective in weight loss if combined with caloric restriction.[29,69]

Exercise is broadly categorized into aerobic and anaerobic. Aerobic, or cardiovascular, exercise builds endurance. It uses O_2 to burn carbohydrates and fats for energy production. Protein and amino acids are used as energy substrates only in cases of starvation, severe caloric deprivation, and overexercising. Anaerobic exercise builds muscle and strength through short bursts of strenuous activity.

Isotonic and isometric exercises are performed during strength training. Isotonic exercise causes a muscle to contract to overcome a resistance and creates movement of the attached joint. It aids in building strength and endurance without overtaxing the cardiovascular system. By contrast, isometric activity involves contraction of the muscle against a fixed resistance and is more stressful on the heart and circulatory system.

Exercise prescription for strengthening should include frequency, intensity, and volume. Strengthening regimens in older adults should focus on multiple muscle groups to avoid overuse and injury of one particular muscle. Strength training is an effective way to improve muscle strength, mass, and neuromuscular activity in older adults, which in turn translates to improved functional capacity while performing everyday activities. It should be performed at least twice weekly, on nonconsecutive days at high to moderate intensity with two to three sets per exercise (set = 8–12 repetitions).[13,18] Maintaining a BMI less than 25 throughout adult life has been recently recommended. For most patients with a BMI between 25 and 30, lifestyle modifications including diet and exercise are appropriate. For individuals over 51 years of age, getting 20 to 30 minutes of moderate activity each day should be combined with daily caloric intake restricted to 1600 for females and 2000 for males. The current *Dietary Guidelines for Americans* recommends limiting sodium intake to 2300 mg (approximately 1 teaspoon) per day for adults; the recommendation is just 1500 mg for those over 51 years of age, as well as Black adults and those with hypertension, diabetes, and kidney disease.[40] Pharmaceuticals should be considered with a BMI greater than 30 or a BMI greater than 27 with an obesity-related disorder. Currently, Tenuate (diethylpropion), Adipex (phentermine), Didrex (benzphetamine), and Bontril (phendimetrazine) are approved for short-term use to provide weight loss. In 2014, the US

Food and Drug Administration approved a new antiobesity medication, Contrave (combination of naltrexone and bupropion), for long-term use. Other medications approved for long-term use include Xenical (orlistat), Belviq (lorcaserin), and Qsymia (phentermine and topiramate extended release).

Pregnancy

Special considerations exist during pregnancy because of the possible competition between exercising maternal muscle and the fetus for blood flow, oxygen delivery, glucose availability, and heat dissipation. Metabolic and cardiorespiratory adaptations to pregnancy can alter the responses from exercise training. The acute physiologic responses to exercise are generally increased during pregnancy compared with pre-pregnancy levels. There are no data in humans to indicate that pregnant females should or should not limit exercise intensity and lower target HRs because of potential adverse effects.[89,137] Healthy, pregnant females without exercise contraindications are encouraged to exercise throughout the pregnancy. Regular exercise during pregnancy provides health and fitness benefits to the mother and child.[32,137] Exercise might also reduce the risk for developing conditions associated with pregnancy, such as pregnancy-induced hypertension and gestational diabetes mellitus.[32,113] For females who have no additional risk factors for adverse maternal or perinatal outcomes, the American College of Obstetricians and Gynecologists (ACOG) has established guidelines for the safe prescription of exercise.[1,2] The Canadian Society for Exercise Physiology's Physical Activity Readiness Medical Examination, or PARmed-X for Pregnancy, should be used for the health screening of pregnant females before their participation in exercise programs.[137] Participation in a wide range of recreational activities appears to be safe during pregnancy. The safety of each sport is determined largely by the specific movements required by that sport. Participation in recreational sports with a high potential for contact, such as ice hockey, soccer, and basketball, could result in trauma to both the female and the fetus. Recreational activities with an increased risk for falling, such as gymnastics, horseback riding, downhill skiing, and vigorous racquet sports, have an inherently high risk for trauma in pregnant and nonpregnant females. Those activities with a high risk for falling or for abdominal trauma should be avoided during pregnancy. Scuba diving should be avoided throughout pregnancy because during this activity the fetus is at increased risk for decompression sickness secondary to the inability of the fetal pulmonary circulation to filter bubble formation. Exertion at altitudes of up to 6000 feet appears to be safe, but engaging in physical activities at higher altitudes carries various risks.

During pregnancy, females can continue to exercise and derive health benefits even from mild to moderate exercise routines. Regular exercise (at least three times per week) is preferable to intermittent activity. Thirty minutes or more of moderate exercise per day on most, if not all, days is recommended. Pregnant females should avoid exercise in the supine position after the first trimester. Such a position is associated with decreased Q in most cases. Because the remaining Q is preferentially distributed away from splanchnic beds (including the uterus) during vigorous exercise, such regimens are best avoided during pregnancy. Prolonged periods of motionless standing should also be avoided. Females should be aware of the decreased oxygen available for aerobic exercise during pregnancy. They should be encouraged to modify the intensity of their exercise according to maternal symptoms and should stop exercising when fatigued and never exercise to exhaustion. Weight-bearing exercises can under some circumstances be continued throughout the pregnancy at intensities similar to those before pregnancy. Nonweight-bearing exercises, such as stationary cycling or swimming, will minimize the risk for injury and facilitate the continuation of exercise during pregnancy.

Exercise should be terminated when any of the following occur: vaginal bleeding, dyspnea before exertion, dizziness, headache, chest pain, muscle weakness, calf pain or swelling, preterm labor, decreased fetal movement, and amniotic fluid leakage. In the case of calf pain and swelling, thrombophlebitis should be ruled out. Pregnant females may participate in a strength-training program that incorporates all major muscle groups, with a resistance that permits multiple repetitions (i.e., 12–15 repetitions) to be performed to a point of moderate fatigue. Isometric muscle actions and the Valsalva maneuver should be avoided, as should the supine position after the first trimester. Morphologic changes in pregnancy should serve as a relative contraindication to types of exercise in which loss of balance could be detrimental to maternal or fetal well-being, especially in the third trimester. Any type of exercise involving the potential for even mild abdominal trauma should be avoided. Pregnancy requires an additional 300 kcal/day to maintain metabolic homeostasis. Females who exercise during pregnancy should be particularly careful to ensure an adequate diet. Those who exercise in the first trimester should augment heat dissipation by ensuring adequate hydration, appropriate clothing, and optimal environmental surroundings during exercise (exercising in cool environments when possible).[137]

The ACOG recommends that females who currently participate in a regular exercise program can continue their training during pregnancy, following the recommendations listed. Studies have demonstrated that females naturally decrease their exercise duration and intensity as their pregnancy advances. Those who begin an exercise program after becoming pregnant are advised to receive physician authorization and begin exercising with low-intensity, low-impact (or nonimpact) activities, such as walking and swimming.[1,2] Contraindications for exercise during pregnancy have also been established by the ACOG (Box 16.7).[2,137] Many of the physiologic and morphologic changes of pregnancy persist 4 to 6 weeks postpartum; therefore the ACSM recommends in general to resume exercise 4 to 6 weeks after delivery. This will vary from one individual to another, with some females able to resume an exercise routine within days of delivery. There are no published studies to indicate that, in the absence of medical complications, rapid resumption of activities will result in adverse effects. Having undergone detraining, resumption of activities should be gradual. No known maternal complications are associated with resumption of training.[137]

Activity for Older Adults

According to the ACSM, it is estimated that the population age 65 years and older will reach 70 million in the United States by 2030. Senior adults age 85+ years will constitute the fastest growing segment of the population. The loss of strength and stamina attributed to aging is in part caused by reduced physical activity. Inactivity usually increases with age. By age 75 years, approximately one in three males and one in two females engage in no exercise. Among adults aged 65 years and older, walking, gardening, and yard work are the most popular physical activities. Social support from family and friends has been consistently and positively related to regular physical activity.[19] Exercise should focus on improving balance, strength, flexibility, and endurance. Senior

• BOX 16.7 Contraindications to Aerobic Exercise During Pregnancy

Absolute Contraindications

- Hemodynamically significant heart disease
- Incompetent cervix, cervical insufficiency, or cerclage
- Intrauterine growth restriction
- Multiple gestation at risk for premature labor
- Persistent second- or third-trimester bleeding
- Placenta previa after 26–28 weeks of gestation
- Premature labor during the current pregnancy
- Restrictive lung disease
- Ruptured membranes
- Severe anemia
- Uncontrolled or poorly controlled hypertension
- Uncontrolled thyroid disease
- Uncontrolled type 1 diabetes
- Unexplained persistent vaginal bleeding, such as in second or third trimester
- Other serious cardiovascular, respiratory, or systemic disorder
- Preeclampsia or pregnancy-induced hypertension

Relative Contraindications

- Anemia or symptomatic anemia
- Cervical dilation
- Chronic bronchitis, mild-moderate respiratory disease, or other respiratory disorders
- Eating disorder
- Extreme morbid obesity
- Heavy smoker
- History of extremely sedentary lifestyle
- History of spontaneous preterm birth, premature labor, miscarriage, or fetal growth restriction
- Malnutrition or extreme underweight
- Mild-moderate cardiovascular disease
- Orthopedic limitations
- Poorly controlled seizure disorder
- Poorly controlled type 1 diabetes
- Recurrent pregnancy loss
- Unevaluated maternal cardiac arrhythmia
- Other significant medical conditions

From Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:187.

adults who are physically active have lower rates of coronary artery disease, hypertension, cerebrovascular accident, type 2 diabetes mellitus, osteoporosis, and a higher level of cardiorespiratory function, muscular strength, and endurance. They demonstrate higher levels of functional health and better cognitive function, and overall have a reduced risk of moderate to severe functional limitations.

Effect of Aging on Muscle

Loss of muscle mass (sarcopenia) results from disuse of the muscle. Aging muscle is also affected by a decrease in growth factors, modifications of motor units, and innervation of the fibers. Although functional motor units decline with age, the surviving axons are required to innervate a higher number of muscle fibers. By age 50 years, total muscle area has shrunk by approximately 10%. The rate of muscle loss accelerates significantly thereafter.[100] Maximal oxygen consumption ($\dot{V}O_2$ max) is used to assess cardiovascular function and maximal capacity. $\dot{V}O_2$ max decreases approximately 5% to 15% per decade starting at 25 to 30 years of age. Similarly, HR_{max} decreases by approximately 6 to 10 beats/min per decade, which contributes to most of the age-related decrease in Q. In older adults, diminishing SV during exercise also causes a drop in HR. Age-related declines in oxidative competence of the muscle and in vascular capacity cause a drop in maximal arteriovenous oxygen difference (the maximum difference in oxygen content between arterial blood and venous blood during exercise) in older adults. These factors, along with diminished oxygen delivery mechanisms and mitochondrial changes, further reduce the capacity to use oxygen at the level of the active skeletal muscle. With exercise training, older adults can attain the same 10% to 30% increase in $\dot{V}O_2$ max as their younger counterparts. This may be attributed to improvements in maximal Q and maximal arteriovenous oxygen difference. The magnitude of change depends on the intensity of the physical activity; low-intensity exercise produces minimal changes.

Exercise is highly beneficial for those with arthritis. Some advantages include muscle strength and flexibility, bone health, and reduced joint pain and fatigue. Improving ROM of the joint and strengthening the surrounding muscles helps decrease the discomfort associated with arthritis during activity.

Aquatic (pool) therapy is particularly helpful for those with both peripheral joint and spinal facet arthropathy. Such a gravity-reduced environment relieves friction and reduces stress on the joints and the spine. Mind-body integrative approaches such as yoga, tai chi, and Pilates are safe and viable alternatives that are thought to lessen inflammation in the body, improve posture, and alleviate stress, although more scientific studies are required.

Age-specific barriers and motivators unique to this cohort are relevant and must be acknowledged. The identification of reliable predictors of exercise adherence will allow health care providers to effectively intervene and change patterns of physical activity in sedentary older adults. In particular, because older patients often respect their physician's advice and have regular contact with their physician, physicians can play a pivotal role in the initiation and maintenance of exercise behavior among the older adult population. Specific recommendations for older adults are outlined in Table 16.5.[12,88] Individualization of resistance training prescriptions

TABLE 16.5 Guidelines for Aerobic Exercise Prescription for Older Adults

FITT	Recommendations
Frequency	≥5 days/week moderate intensity OR ≥3 days/week vigorous intensity OR 3–5 days/week of combination moderate and vigorous intensity
Intensity	On a scale of 0–10 for level of physical exertion, 5–6/10 for moderate intensity and 7–8/10 for vigorous intensity
Time	For moderate intensity, 30–60 min/day. For vigorous intensity, 20–30 min/day. For combination moderate and vigorous intensity, an equivalent time combination. Does not need to be all at once, may be accumulated over the course of the day.
Type	Any modality that does not impose excessive orthopedic stress, such as walking, aquatic exercise, and stationary cycle may be advantageous for those with limited tolerance for weight-bearing activity.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:183.

is also essential and should be based on the health and fitness status and specific goals of the participant. Some guidelines follow, with reference to the intensity, frequency, and duration of exercise (Table 16.6).[88,137]

Flexibility Training

Static stretching is considered safer than dynamic stretching for older adults (Box 16.8). The American Heart Association recommends holding each stretch for 10 to 30 seconds, with three to four repetitions.[106] Older individuals may benefit from holding a stretch for 30 to 60 seconds.

Resistance Training. Regardless of which specific protocol is adopted, several commonsense guidelines pertaining to resistance training for older adults should be followed.[88,137] The major goal of the resistance training program is to develop sufficient muscular fitness to enhance an individual's ability to live a physically independent lifestyle. The first several resistance training sessions should be closely supervised and monitored by trained personnel who are sensitive to the special needs and capabilities of older adults. Begin (the first 8 weeks) with minimal resistance to allow for adaptations of the connective tissue elements. Teach proper training techniques for all the exercises to be used in the program. Instruct older participants to maintain their normal breathing pattern while exercising. As a training effect occurs, achieve an overload initially by increasing the number of repetitions and then by increasing the resistance. Never use a resistance that is so heavy that the exerciser cannot perform at least eight repetitions. Stress that all exercises should be performed in a manner in which the speed is controlled (no ballistic movements should be allowed). Perform the exercises in a ROM that is within a pain-free arc (i.e., the maximal range of motion that does not elicit pain or discomfort). Perform multiple-joint exercises (as opposed to single-joint exercises). Given a choice, use machines to resistance train as opposed to free weights (machines require less skill to use, protect the back by stabilizing the user's body position, and allow the user to start with lower resistances, to increase by smaller increments, and to more easily control the exercise ROM). Heavy free weights should be used only by those who have had special training in how to lift properly and who have a spotter with them during the exercise. Do not overstrain. Two strength-training sessions per week are the minimum number required to produce positive physiologic adaptations. Depending on the circumstances, more sessions might not be productive. Participants with arthritis should never participate in strength-training exercises during active periods of joint pain or inflammation. Engage in a year-round resistance-training program on a regular basis. When returning from a layoff, start with resistances of less than 50% of the intensity at which the participant had been previously training (as tolerated), then gradually increase the resistance. Flexibility exercises should be performed on major muscle groups with prolonged (not ballistic) stretching at an intensity of 50% to 60% maximal at least 2 days per week. Balance exercises should be incorporated for frequent fallers or those with mobility problems.

TABLE 16.6 Guidelines for Resistance Exercise Prescription for Older Adults

FITT	Recommendations
Frequency	≥2 days/week
Intensity	Progressive weight training: light intensity (40%–50% 1-rep-max for beginners); progress to moderate-to-vigorous intensity (60%–80% 1-rep-max); alternatively, moderate (5–6) to vigorous (7–8) intensity on a 0–10 scale. Power training: light-to-moderate loading (30%–60% of 1-rep-max).
Time	Progressive weight training: 8–10 exercises involving the major muscle groups; >1 set of 10–15 repetitions for beginners; progress to 1–3 sets of 8–12 repetitions for each exercise. Power training: 6–10 repetitions with high velocity.
Type	Progressive or power weight-training programs or weight-bearing calisthenics, stair climbing, and other strengthening activities that use the major muscle groups.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:183.

• BOX 16.8 Physical Activity Counseling for Older Adults: Quick-Guide Recommendations

Aerobic
- ≥30 min or 3 sessions of ≥10 min/day
- ≥5 days/week
- Moderate intensity = 5–6 on a 10-point scale (where 0 = sitting, 5–6 = "can talk," and 10 = all-out effort)
- In addition to routine activities of daily living.

Strength
- 8–10 exercises (major muscle groups), 10–15 repetitions
- ≥2 nonconsecutive days/week
- Moderate to high intensity = 5–8 on a 10-point scale (where 5–6 = "can talk" and 7–8 = shortness of breath)

Flexibility/Balance
- ≥10 min ≥2 days/week
- Flexibility to maintain/improve range of motion (i.e., stretching of major muscle-tendon groups, yoga)
- Balance exercises for those at risk for falls (i.e., tai chi, individualized balance exercises)

Prevention
- Create a single physical activity plan that integrates preventive and therapeutic treatment of chronic conditions

Children

Children tend to be more active than adults and accordingly tend to maintain adequate levels of physical fitness. Healthy children should be encouraged, nonetheless, to engage in physical activity on a regular basis. However, because children are anatomically, physiologically, and psychologically immature, special precautions should be applied when designing exercise programs. Children can experience a higher incidence of overuse injuries, or they damage the epiphyseal growth plates if endurance exercise is excessive. The risk for injury can be significantly decreased by ensuring appropriate matching of competition in terms of size, maturation, or skill level; the use of properly fitted protective equipment; and liberal adaptation of rules toward safety, proper conditioning, and appropriate skill development. Children have less efficient thermoregulation than that of adults and are more prone to hyperthermia and hypothermia.[88,137] The current rise in childhood obesity underscores the importance of regular exercise. In the United States, 32% of children are overweight or obese.[95,109] This increase in obesity has been linked to increases in comorbidities, including glucose intolerance, type 2 diabetes, hypertension, and hyperlipidemia.[95] Studies have demonstrated that monitored programs of moderate to vigorous exercise can result in a decrease in percent body fat and improvement of insulin resistance.[101] Consensus guidelines for 2005 recommend that schools provide for 30 to 34 minutes of daily vigorous activity.[129] The Endocrine Society recommends 60 minutes of daily vigorous activity.[4]

Specific considerations for children include[137]:

- Children and adolescents may safely participate in strength-training activities after proper instruction and supervision. Eight to 15 repetitions of an exercise should be performed to the point of moderate fatigue with good mechanical form before the resistance is increased.
- Because of immature thermoregulatory systems, youth should exercise in thermoneutral environments and be properly hydrated.
- Children and adolescents who are overweight or physically inactive may not be able to achieve 60 min/day of physical activity. Therefore gradually increase the frequency and time of physical activity to achieve this goal.
- Efforts should be made to decrease sedentary activities (i.e., television and video games) and increase activities that promote lifelong activity and fitness (i.e., walking and cycling).

ACSM guidelines for exercise prescription in children are detailed in Table 16.7.[137]

TABLE 16.7 Guidelines for Strength Training in Children and Adolescents

FITT	Recommendations
Frequency	Aerobic: daily, with vigorous intensity 3 day/week Resistance: At least 3 day/week Bone strengthening: At least 3 day/week
Intensity	Aerobic: moderate (noticeable increase in heart rate) to vigorous intensity (substantial increase in heart rate and breathing). Resistance: Use of body weight as resistance of 8–15 submaximal repetitions of an exercise to the point of moderate fatigue with good mechanical form. Bone strengthening: variable with emphasis on activities that produce moderate to high bone loading through impact or muscle force production.
Time	As a part of ≥60 min/day exercise (for each aerobic, resistance, and bone strengthening exercise).
Type	Aerobic: enjoyable and developmentally appropriate activities, including tag/running games, hiking/brisk walking, hopping, skipping, jumping rope, swimming, dancing, bicycling, and sports such as soccer, basketball, or tennis. Resistance: can be unstructured (playing on playground equipment, climbing trees, tug of war), or structured and appropriately supervised (body weight exercises such as pushups and situps, lifting weights, or working with resistance bands). Bone strengthening: examples include running, jumping rope, basketball, tennis, resistance training, and hopscotch.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:169.

Hypertension

The ACSM makes the following recommendations regarding exercise testing and training of individuals with hypertension.[88,137] Mass exercise testing is not advocated to determine those individuals at high risk for developing hypertension in the future as a result of an exaggerated blood pressure response. However, if exercise test results are available and an individual has an exercise blood pressure response above the 85th percentile, this information provides some indication of risk stratification for that patient and the necessity for appropriate lifestyle behavior counseling to ameliorate this increase. Endurance exercise training by individuals who are at high risk for developing hypertension can reduce the rise in blood pressure that occurs with time, justifying its use as a nonpharmacologic strategy to reduce the incidence of hypertension in susceptible individuals. Endurance exercise training elicits an average reduction of 10 mm Hg for both SBP and DBP in individuals with mild essential hypertension (blood pressures in the range of 140/90–180/105 mm Hg) and secondary hypertension resulting from renal dysfunction. The recommended mode, frequency, duration, and intensity of exercise are generally the same as those for apparently healthy individuals. Exercise training at somewhat lower intensities (e.g., 40%–70% of $\dot{V}O_2$ max) appears to lower blood pressure as much, or more, than exercise at higher intensities. This can be especially important in specific hypertensive populations, such as older adults. Based on the high number of exercise-related health benefits and low risk for morbidity and mortality, it seems reasonable to recommend exercise as part of the initial treatment strategy for individuals with mild to moderate essential hypertension. Individuals with marked elevations in blood pressure should add endurance exercise training to their treatment regimen only after a physician's evaluation and the initiation of pharmacologic therapy. Exercise can reduce their blood pressure further, allowing them to decrease their antihypertensive medications and attenuate their risk for premature mortality. Resistance training is not recommended as the primary form of exercise training for individuals with hypertension. With the exception of circuit weight training, resistance training has not consistently been shown to lower blood pressure. Resistance training is recommended as a component of a well-rounded fitness program, but it should not be the only form of exercise in the program. If resting SBP is greater than 200 mm Hg and/or DBP is greater than 110 mm Hg, then exercise is contraindicated. When exercising, it appears prudent to maintain an SBP of 220 mm Hg or less and/or a DBP of 105 mm Hg or less. Specific guidelines for exercise in patients with hypertension are listed in Table 16.8.[88,137]

Peripheral Vascular Disease

Patients with peripheral vascular disease experience ischemic pain (claudication) during physical activity as a result of a mismatch between active muscle oxygen supply and demand. The symptoms can be described as burning, searing, aching, tightness, or cramping. Pain is most often experienced in the calf, but it can begin in the buttock region and radiate down the leg. The symptoms typically disappear on cessation of exercise, although some

TABLE 16.8 Guidelines for Exercise Prescription in Patients With Hypertension

FITT	Recommendations
Frequency	Aerobic: 5-7 days/week Resistance: at least 2 days/week Flexibility: at least 2 days/week
Intensity	Aerobic: moderate (40%–60% oxygen uptake reserve or heart rate reserve; rating of perceived exertion 12–13 on a scale of 6–20 points). Resistance: moderate (60%–70% of 1-rep-max; may progress to 80% 1-rep-max; for older individuals and novice exercisers, begin with 40%–50% 1-rep-max). Flexibility: stretch to the point of feeling tightness or slight discomfort.
Time	Aerobic: at least 30 min/day (continuous or accumulated). Resistance: 2–4 sets of 8–12 repetitions of each of the major muscle groups per session to total at least 20 min/session with rest days interspersed depending on the muscle groups being exercised. Flexibility: hold static stretch for 10–30 seconds with 2–4 repetitions of each exercise targeting the major muscle-tendon units to 60 seconds of total stretching time for each exercise; ≤10 min/session
Type	Aerobic: prolonged, rhythmic activities using large muscle groups (e.g., walking, cycling, and swimming). Resistance: resistance machines, free weights, resistance bands, and/or functional body weight exercise. Flexibility: static, dynamic, and/or proprioceptive neuromuscular facilitation.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:290.

TABLE 16.9 Guidelines for Exercise Prescription in Patients With Peripheral Arterial Disease

FITT	Recommendations
Frequency	Aerobic: 3–5 days/week Resistance: at least 2 (nonconsecutive) days/week Flexibility: at least 2 day/week, with daily being most effective
Intensity	Aerobic: moderate (i.e., 40%–60% oxygen uptake reserve) to the point of moderate pain (3 of 4 on the claudication pain scale) or from 50%–80% of maximum walking speed. Resistance: 60%–80% of 1-rep-max. Flexibility: stretch to the point of feeling slight discomfort or tightness.
Time	Aerobic: 30–45 min/day for minimum of 12 weeks, then may progress to 60 min/day. Resistance: 2–3 sets of 8–12 repetitions; 6–8 exercises targeting the major muscle groups. Flexibility: 10- to 30-second hold for static stretching; 2–4 repetitions of each exercise.
Type	Aerobic: weight-bearing (i.e., free or treadmill walking) intermittent exercise with seated rest when moderate pain is reached and resumption when pain is *completely* alleviated. Resistance: whole body, focusing on large muscle groups; emphasis on lower limbs if time limited. Flexibility: static, dynamic, and/or proprioceptive neuromuscular facilitation stretching.

Adapted from Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:247.

patients can have claudication at rest in severe cases. Severe peripheral vascular disease is treated initially with exercise and medications that decrease blood viscosity. Treatment with angioplasty or bypass grafting might also be indicated. Weight-bearing exercise is preferred to facilitate greater functional changes, but it might not be well tolerated initially. Prescription of nonweight-bearing exercise (which can permit a greater intensity or longer duration) is a suitable alternative.[88,137] Specific guidelines for exercise in patients with peripheral vascular disease are listed in Table 16.9.[88,137]

Diabetes

The response to exercise in the patient with type 1 diabetes mellitus depends on a variety of factors, including the adequacy of control by exogenous insulin. If the patient is under appropriate control or only slightly hyperglycemic without ketosis, exercise can decrease blood glucose concentration and lower the insulin dose required. Patients with type 1 diabetes mellitus must be under adequate control before beginning an exercise program. Serum glucose concentrations in the general range of 200 to 400 mg% (mg/dL) require medical supervision during exercise, and exercise is contraindicated for those with fasting serum values greater than 400 mg%. Exercised-induced hypoglycemia is the most common problem experienced by exercising patients with diabetes.

Hypoglycemia can occur not only during the exercise but for up to 4 to 6 hours after an exercise session.[88,137] The risk for hypoglycemic events can be minimized by taking the following precautions:

- Monitor blood glucose frequently when initiating an exercise program.
- Decrease insulin dose (by 1–2 units as prescribed by the physician) or increase carbohydrate intake (10–15 g carbohydrate per 30 minutes of exercise) before an exercise session.
- Inject insulin in an area such as the abdomen that is relatively inactive during exercise.
- Avoid exercise during periods of peak insulin activity.
- Eat carbohydrate snacks before and during prolonged exercise sessions.
- Be knowledgeable of the signs and symptoms of hypoglycemia and hyperglycemia.
- Exercise with a partner.

Other precautions that should be taken include the following[88,137]:

- Use proper footwear and practice good foot hygiene.
- Be aware that beta blockers and other medications can interfere with the patient's ability to discern hypoglycemic symptoms and/or angina.
- Be aware that exercise in excessive heat can cause problems in diabetic patients with peripheral neuropathy.
- Patients with advanced retinopathy should not perform activities that cause excessive jarring or marked increases in blood pressure.

TABLE 16.10 Guidelines for Exercise Prescription in Patients With Diabetes

Component	Details
Frequency	Aerobic: 3–7 days/week, with no more than 2 consecutive days without activity Resistance: minimum of 2 nonconsecutive days per week Flexibility and balance: at least 2 days/week
Intensity	Aerobic: moderate to vigorous (based on subjective experience). Resistance: moderate (50%–70% of 1-rep-max) to vigorous (70%–85% 1-rep-max). Flexibility and balance: stretch to the point of tightness or slight discomfort; light to moderate for balance exercises.
Time	Aerobic: 150 min/week at moderate-vigorous intensity. Resistance: at least 8–10 exercises with 1–3 sets of 10–15 repetitions to near fatigue per set early in training. Flexibility and balance: hold static stretch for 10–30 seconds; 2–4 repetitions of each exercise. Balance for any duration.
Type	Aerobic: prolonged, rhythmic activities using large muscle groups (e.g., walking, cycling, swimming, or high-intensity interval training). Resistance: resistance machines, free weights, resistance bands, and/or body weight. Flexibility and balance: static, dynamic, yoga, other stretching.

Adapted Liguori G, Feito Y, Fountaine CJ, et al., eds. *ACSM's Guidelines for Exercise Testing and Prescription.* 11th ed. Wolters Kluwer; 2021:280.

- Patients should have physician approval to resume exercise training after laser treatment.

Specific guidelines for exercise in patients with diabetes are listed in Table 16.10.[88,137]

Myofascial Pain Syndrome and Fibromyalgia

Chronic musculoskeletal pain of various etiologies affects many adults. It is characterized by fascial contraction and trigger points in the muscles, causing pain and decreased ROM. This adversely affects quality of life. Patients may present with a knot or fibrous band in the muscle, most commonly located in the upper traps, rhomboids, and paraspinals. There are multiple interventions for myofascial pain syndrome, including nonsteroidal antiinflammatory drugs and trigger point injections; however, massage, stretching, and activity are most effective in the management and prevention of myofascial pain.

The prevalence of fibromyalgia is approximately 2% to 4%, most commonly affecting females. It is characterized by widespread muscular pain, headache, fatigue, sleep disturbances, and memory/cognitive problems. Working females with fibromyalgia who are hospitalized for chronic musculoskeletal issues were almost 10 times less likely to return to work and 4 times less likely to retain work 1 year posthospitalization. Low-intensity aerobic exercise combined with gentle stretching should be prescribed for patients with fibromyalgia. The goal is to maintain function in day-to-day activities. Pain is the most common limiting factor. Clinical trials have shown that aerobic exercise decreases pain levels, thereby improving function and quality of life. One particular study trained patients for 25 min/day two to three times per week with an average intensity of 50% $\dot{V}O_2$ max. Unlike the control group, the testing group demonstrated a decrease in the number of tender points and painful body surface area. A study published by the American College of Rheumatology suggests that there is a linear relationship between the number of steps taken daily and relief from symptoms of fibromyalgia. It is recommended that patients gradually accumulate at least 5000 additional steps daily.[85] McLoughlin et al.[103] proved that using functional magnetic resonance imaging that increased physical activity was associated with decreased perception of pain in females with fibromyalgia. Patient education and cognitive therapy are also important aspects in the management of fibromyalgia. Patients should be counseled on how to prioritize their time to achieve a balance among work, daily responsibilities, and leisure.

Acupuncture is increasing in popularity to alleviate the symptoms; however, data supporting this is sparse, and further investigations are required. Other modalities such as yoga, tai chi, massage, deep breathing, and meditation therapy are widely employed.[12]

Organ Transplantation

After cardiac transplantation, abnormal aerobic tolerance is caused by deconditioning before transplant, chronotropic incompetence from denervation, skeletal muscle weakness, and immunosuppressive medications (steroids, cyclosporine). Long-term use of cyclosporine causes muscle atrophy and a shift toward a larger amount of fast-twitch muscle fibers at the expense of the slow-twitch fibers, whereas corticosteroids result in mitochondrial dysfunction. A study of patients with a left ventricular assist device and cardiac transplant demonstrated improved quality of life and physical activity within the first 3 months of surgery but stayed relatively unchanged thereafter and always remained well below their healthy counterparts.[79] The importance of cardiac rehabilitation was illustrated in a study that randomized 27 patients within 2 weeks of cardiac transplantation to a 6-month structured rehabilitation program or unstructured therapy at home. Compared with the control group, the exercise group had a greater increase in peak oxygen consumption (49% vs. 18%) and workload (59% vs. 18%) and a greater reduction in the ventilatory equivalent for carbon dioxide (20% vs. 11%).[112] As a result of persistent sinus node denervation, HR cannot be used as a measure of work intensity. Most commonly, the Borg scale is used to quantify the intensity of the activity being performed. Patients with an organ transplant are deconditioned and weak postoperatively. Although data are limited, rehabilitation after solid organ transplantation appears highly beneficial. A study of patients with renal transplant showed that rehabilitation posttransplant is helpful in aiding psychological and physical recovery, thereby improving quality of life.[146]

Summary

The benefits of exercise for health and human performance are well known. Applying appropriate exercise prescriptions based on the physiologic response to exercises and the principle of specificity of training will ensure an appropriate training response and minimize the risk for injury. Special populations, including the young, older adults, and those with disease states, may require specific modifications in exercise programs to maximize safety.

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